FONTAINE_DECAL Project – Final Design Package – Senior Design II

Date	Revision	Author	Comments
2026-05-01	1	FONTAINE_DECAL Team	Original Draft

Table of Contents

1 Overview of this Document

This document describes the design, implementation, and validation of the FONTAINE_DECAL project and its final product. Fontaine Modification currently applies customer decals through a manual and labor-intensive process that requires technicians to measure each decal location by referencing layout documents provided by OEMs. To improve efficiency and reduce human error, Fontaine Modification sought to replace these manual measurements with an automated projection system capable of displaying accurate and properly scaled decal layouts directly onto truck doors. This document provides an overview of the system design, hardware and software integration, testing and validation results, and system performance. It also summarizes communication between the FONTAINE_DECAL team, faculty mentor, and Fontaine Modification. Caleb Burton, identified as the Project Lead, is responsible for maintaining this document.

2 Project Overview/Statement of Work Summary

The objective of this project was to remove the manual decal measurement process by designing an automatic projection system that uses computer vision to digitally project an arrangement of decals onto truck doors. The solution utilizes a mobile hardware suite comprising a camera, a microcomputer, and a projector to detect truck door geometry in real time. By employing feature extraction algorithms, the system computes precise spatial coordinates and overlays digital layouts directly onto the vehicle. This provides operators with immediate, high-accuracy visual references for marking and application while adhering to all safety standards and performance specifications. The resulting prototype streamlines the assembly workflow and ensures consistent quality for high-volume production by eliminating manual measurement variability. Furthermore, the inclusion of comprehensive user manuals and assembly drawings ensures the system is easily reproducible for future deployment.

3 Design Narrative

The deliverables for this project include a mobile prototype of a projection system that can automatically identify a truck door and project a specified decal layout. Functionality of the system is dependent on the placement of the truck within the agreed upon tolerances defined in the Statement of Work.

The system is designed to work in a cyclical manner, projecting the same decal layout onto a number of identical trucks, with each cycle beginning with the operator's placement of the truck and triggering of the projection. Should the prototype prove to be an effective part of the support's high volume workflow, the system will be adapted to a permanent ceiling mount. Development of such a permanent mount is outside the scope of this project. All components of the projection system should be readily available for procurement or replacement while minimizing the amount of custom fabrication required so that it is possible in the future for a large quantity of units to be deployed to multiple locations.

Based on the criteria described above we determined the best and most cost effective solution is to purchase a preexisting cart with a large post and modify it to suit the project. Atop this post will sit three components that comprise the sensing and projecting head: a high resolution camera, a high resolution laser projector, and an AI powered edge computer. The system will operate at a distance of 8 feet from the truck door, while the selected parts allow for flexibility on this working distance in future applications.

The camera selection is what drove the design of the rest of the system. High quality input is required to maintain a reliable and usable output image. We selected the e-con Systems See3CAM_CU135M, a 13MP (4208x3120px) camera with a narrow (56°x43°) field-of-view (FOV) to maximize the pixel density within our region of interest (ROI) on the door. The higher resolution allows for finer levels of control at every subsequent step of the system. In practice, this means that sharper object segmentation allows for us to construct more accurate axes for localizing the projections, and more accurately detect the 3D contour of the door and warp the projections accordingly. To ensure the high quality image input data is usable, the camera's focal point has been taken into consideration when mounting. This means that there is a known offset in a single dimension between the camera's and the projector's focal points allowing for the simplest adjustment between them during image processing.

Figure 1. e-con Systems See3CAM_CU135M

The ROI for the projector is significantly smaller than that of the camera, which allows for more flexibility with selection. We opted for the ViewSonic LS740HD which has a resolution of 1920x1080p and a maximum optical zoom of 1.3x. Even with the lower resolution in comparison to the camera, the effective pixel density is comparable since the ROI is so much smaller and the project has a narrower FOV of 39°x22°. Additionally, this projector was selected due to its laser technology which provides a high brightness of 5,000 lumens at an expected lifetime of 20,000 hours. With a contrast ratio of 3,000,000:1 the projector is capable of displaying sharp lines which enable the operator to clearly mark any reference points by hand with a dry erase marker.

Figure 2. ViewSonic LS740HD

For the final main components, we selected the NVIDIA Jetson Orin Nano Super Developer Kit to perform all the processing for the system. This model of the Jetson runs a Linux operating system and with 8GB of memory it is capable of running large neural networks at real time speeds. The Jetson will be the small edge-computer running our custom Python scripts and facilitating connectivity with our other devices. The HDMI port and RJ45 ethernet port will allow the Jetson to send images and commands to the projector while one of the five USB ports will be dedicated to receiving input from the camera. The Jetson will be mounted inside of an aluminum enclosure that protects the PCB and still allows access to the I/O ports. A dedicated housing case for the Jetson was purchased to provide additional security.

Figure 3. NVIDIA Jetson Orin Nano (Super Developer Kit)

The core components of the system are all mounted to a single aluminum plate that is fixed to the top of the mobile post. This design is to mimic the overhead ceiling mounted position that is common for projectors, because this is the intended permanent position for the system should it be adopted as a part of the decal application workflow. This over-the-shoulder projection, as opposed to from the ground or from the side, reduces the amount of the image that is obscured by the operator and poses the smallest risk as a tripping hazard. The frame of the mount is adjustable with locking handles to provide a range of possible projection angles should it be necessary to change the projection distance. A small amount of adjustment is also necessary to compensate for projection offset tolerances caused by manufacturing differences in the projection lens. One area of refinement was in the design of camera brackets. An 8° downward tilt had to ensure the camera’s field of view captured the entire ROI of the truck door.

Figure 4. Sensing Head

Figure 5. Mobile Stand

The software that enables our system to operate is composed of a few distinct parts that all allow for a corrected decal to be projected for any truck positioned and oriented according to the project tolerances. All software is run locally on the Jetson, made up of Python scripts that utilize the OpenCV library as well as a custom trained version of the YOLOv8 object recognition and image segmentation model. From the image input the door is segmented and using a provided 3D CAD model of the door enabling keystone correction of the decals. A digital coordinate system is then applied to the segmentation to place the decals in the correct position according to the decal layout document unique to each job order. This digital coordinate system and decal alignment utilize mathematical models to solve for planar homography and pose estimation. These equations allow the software to map the 2D decal coordinates onto the 3D geometry of the door surface, ensuring accurate spatial orientation and perspective correction. These equations can be seen in further detail in Appendix C. Finally, the localized and corrected decal image is then sent to the projected and displayed for the operator to apply the decals using the new visual aid. The operator will be able to wirelessly enable or disable the projection when not in use. The operator interfaces with a streamlined, user-friendly GUI featuring a dedicated status bar that provides real-time feedback on the projection system's current process stage.

Figure 6. YOLOv8 Segmentation Mask

Figure 7. Decal Projection

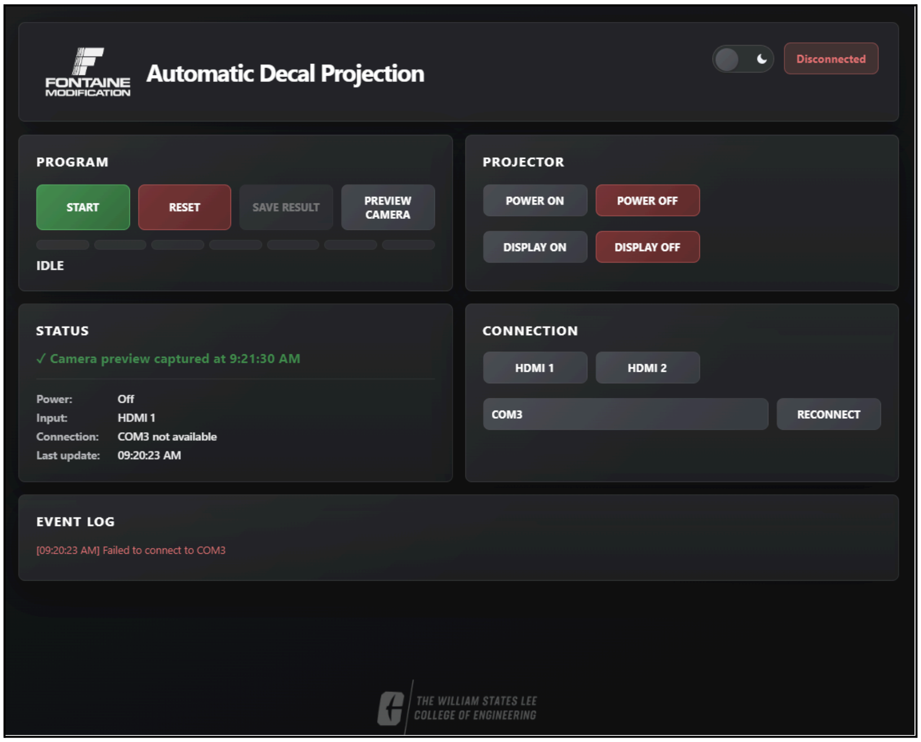


Figure 8. User Interface Dashboard

The system is powered by an integrated power strip that is plugged into a standard 120V AC outlet. From this power strip that has multiple outlets the Jetson and projector will be plugged in, where the Jetson will draw a maximum of 25W, the projector a 210W or 0.5W when in standby, and the camera will draw 2W from its USB-C connection to the Jetson. This gives the entire system a maximum power draw of 237W.

The production of our software followed the team's projected plan including some of our predicted challenges from SD1. Such challenges being door detection and decal alignment, both of which were highly dependent on one another. If the door detection system misidentified a segment of the door, the projected decal would be skewed or misplaced. Such errors would fall outside our given performance specifications. To fix this issue, we collected a large dataset of truck imagery both in the lab and onsite. This allowed us to increase the door directions systems precision.

4 Test Results

The automated decal projection system was tested in the Industrial Solutions Laboratory throughout the semester. While many manual-placing experiments took place, the first official automated decal layout validation test was conducted in the ISL on April 7. This test included the projection system attempting to project all four decals on the blank door within the required quarter-inch tolerance of the layout document dimensions. The results exhibited inconsistencies in clock rotation and placement. On April 17, the team took an improved system to the Fontaine Statesville location for an extensive test and training session. The automated precision was tested and gave consistent failures. These failures consisted of the decals being too high on the door, leading to surpassing the quarter-inch tolerance of decal placement. In a similar fashion, there were errors in the decal central column being clocked too far “counter-clockwise” or “clockwise”. On extreme deviations, the decals would be placed over six inches out of layout bounds and incorrectly warped. These tests and training gave a direct insight into the significant differences in lab trials against real truck environments. The segmentation mask that was trained particularly for this truck was found to be the cause of the on-site error. Over 150 training images were taken on-site in order to train the AI model for detecting precise door landmarks.

More tests followed in the ISL lab with great success once a new segmentation model was trained. 90% of the tests concluded in automated detection of the door, leading to correct warping and placement of the projected decals within 15 seconds of the process being triggered via the wirelessly connected GUI. With this progress, the team initiated another on-site validation session at Fontaine Modification in Statesville, NC on April 27 (accompanied by the team mentor Michelle Demers). This validation consisted of over 50 projection runs with varying projection distances and angles. The results were inconsistent again. The debugging programs that the team had created for troubleshooting showed that the new segmentation model was good enough to use, however the line detection was not built with enough leniency to handle the new model’s edges. April 28, the team gathered to make needed changes to the software while the decal placement projector stayed in Fontaine’s possession; running multiple tests for each cloud-updated software change the team gave to the supporters. This was the first time the system was able to automatically align to the real truck door and place decals within SOW specifications.

On April 30, the team sent select members to the Fontaine Modification site to do a final validation. The system did produce program iterations that did not successfully place decals within the quarter-inch tolerance, but the supporter was pleased to see that within the testing, most of the runs that incorporated an intentional placement of the stand, the decals were automatically placed with near perfect accuracy. The projection allowed the supporter to mark fiducials on the door to place real decal stickers or simply align the decal with the projection for confidence. Leaving the impression that when the user has intentionally set up the stand and door arrangement, the projection will be >95% accurate to the layout sheet. Trial images of the decal projections and associated segmentation masks for the specified test dates are documented in Appendix D.

5 Adherence to Engineering Standards and Codes

The prototype has been carefully designed to comply with established engineering standards that ensure both safety and reliability throughout its intended use. Occupational Safety and Health Administration (OSHA) guidelines are followed to maintain a safe workplace by properly identifying and controlling potential hazards such as electrical risks, trip hazards, and ergonomic concerns during assembly, operation, and maintenance. In addition, the International Electrotechnical Commission (IEC) standards are observed to guarantee that the device meets globally recognized safety requirements for electrical and electronic systems, with particular emphasis on low-voltage components and power infrastructure. To further ensure regulatory compliance within the United States, the design also adheres to the National Electrical Code (NFPA 70), which provides rigorous requirements for safe electrical installation and protection against fire or electrocution. Additionally, all technical drawings adhere to ASME Y14.5 standards for Geometric Dimensioning and Tolerancing (GD&T), providing a standardized framework for part fabrication and inspection.

A thorough and structured risk assessment was conducted to identify hazards associated with the device, along with corresponding mitigation strategies to minimize user exposure. To address the risk of electrical shock resulting from damaged or exposed cables, all connections utilize properly insulated wiring and are supported by clearly marked warning labels to promote safe handling. Risks of falling or crushing injuries, including potential pinch points, are mitigated through the design of a stable base that is properly rated for the device’s total weight and height. Mechanical guards are incorporated where necessary, and the tipping moment about the base has been evaluated to ensure sufficient stability under normal and expected operating conditions.

Tripping hazards posed by power cables or the cart base are reduced through dedicated cable-management features and clearly marked risk zones that help guide users around high-traffic areas. To prevent visual disturbance, discomfort, or eye strain, the system includes clear user instructions for safe projector operation, emphasizing proper positioning and avoidance of direct exposure to the light beam. The projector is also equipped with an automatic standby mode to ensure light is only emitted when required. Finally, fire and overheating risks are minimized by selecting a projector with built-in cooling systems and thermal cut-off protection. The device is intended for intermittent operation, allowing proper cooling intervals that prevent prolonged heat buildup and contribute to long-term operational safety and performance.

6 Evaluation of Prototype as Compared to Project Performance Specification Document

The hardware of this system meets all performance specifications. Both in its physical limitations and projection outputs. All hardware falls within the expected environmental conditions such as temperature, humidity, and lighting. The mobile post was fabricated with OSHA, IEC, NFPA, and GD&T standards. The sensing head itself weighs less than 100lbs and the mobile post is within the 60” x 60” x 96” footprint. On site test data confirms that the decal projection also meets all performance specifications. All projected layouts were within ±0.25 inches. This projection of the decals were visible across both lab and shop lighting conditions. Final software adjustments were needed based on additional onsite training. This allowed the system to be fully functional within the Fontaine modification facility.

7 Recommendations for Further Development

While the system successfully demonstrates automated decal projection and placement, several opportunities exist to improve system robustness, scalability, and deployment readiness.

Expanded Segmentation Model – Future development should include training segmentation models on a broader range of truck models, cab geometries, and surface conditions. While the current model performs well on the Cascadia P4, additional datasets would improve generalization across different truck manufacturers, door geometries, paint colors, lighting conditions in production environments. To improve training efficiency, techniques such as knowledge distillation can be leveraged to transfer learned features into smaller, faster models suitable for real-time deployment without loss in segmentation mask accuracy.

Multi-Layout Decal Configuration – The current system is designed around a single decal layout configuration. Future iterations should support: multiple decal layouts selectable through the user interface, dynamic scaling and positioning for different truck models, integration with OEM layout files or CAD data. This would allow the system to adapt to varying customer specifications without requiring code changes.

Flexible Sensing Head Deployment – The current system uses a mobile mounting platform for positioning the camera and projector. Future development should explore alternative mounting configurations, such as overhead or ceiling-mounted systems, to reduce floor space usage and improve integration within existing shop environments. An elevated sensing head could provide a more consistent viewing angle, minimize any potential obstruction from operators or equipment, and reduce variability introduced by manual repositioning of the mobile stand. This would improve repeatability and streamline decal projection workflows.

These advancements would support broader deployment and enable the system to meet the demands of high-volume industrial applications.

8 Impact

The impact of the system on public welfare is negligible, as the prototype and any future units will be confined to a stationary role in a controlled warehouse environment. In the direct sense, the impact of the system for the operator and supporter company is largely positive. It has been fabricated with the environment in mind, ensuring the system is reusable with a long lifespan and minimal amount of parts requiring maintenance or replacement. The design is economical as it allows for faster cycle time on decal applications with short intuitive training, which enables for a global scalable application with consistent results. With operator confidence being key in the design, the system allows for development of workplace culture and reducing frustration from monotonous tasks and increasing social fulfillment.

9 Bill Of Materials (BOM)

A total of $2,783.95 of our allotted $3,250 budget was spent on the fabrication of the mobile post. This expenditure covered material stock, lumber and fasteners. This cost does not reflect the total build cost of the prototype. The total build cost includes all Global Industrial parts which were purchased and provided to us by Fontaine Modification. That cost also includes any excess ISL material used or purchased items by the team. The bill of materials for this project can be seen in Figure 9. All purchase requisites are provided in Appendix E.

Name	ID	Source	Subsystem	Part Code & URL	Qty Used	Quantity per Pack	Price	Unit Price	Cost	Purcha
ViewSonic LS740HD 1080p Laser Projector	1	Amazon	Electrical	B0CDG3JKQP	1	1	$1,069.02	$1,069.02	$1,069.02	ISL
NVIDIA Jetson Orin Nano Dev Kit	2	Amazon	Electrical	B0DNZC8JGZ	1	1	$399.00	$399.00	$399.00	ISL
NVIDIA Jetson Enclosure	3	Amazon	Electrical	B0CG3BBS5S	1	1	$22.99	$22.99	$22.99	ISL
Din Rail Mount Clip Snap	4	Amazon	Mount	B0B7VB7SVK	1	4	$7.99	$1.99	$7.99	ISL
Multi Purpose Rubber Spacer	5	Amazon	Mount	B0DQY4Y9B3	2	18	$6.99	$0.38	$6.99	ISL
HDMI to DisplayPort cable 6ft	6	Amazon	Electrical	B0BV2KKX2R	1	1	$6.79	$6.79	$6.79	ISL
Cat 8 Eithernet Cable	7	Amazon	Electrical	B08PL53JG3	1	1	$3.99	$3.99	$3.99	ISL
25ft Extension cord with 6 outlets	8	Amazon	Electrical	B0CGVB559P	1	1	$21.58	$21.58	$21.58	ISL
AGVEE 2 Pack Mid Angle USB4 USB-C Male to USB-C Female Ada	9	Amazon	Electrical	B0DQXNB3DN	1	1	$8.99	$8.99	$8.99	ISL
Under Desk Cable Management- 2 Pack Cord Organizer. Metal Wire	10	Amazon	Mount	B08KFT479X	1	2	$24.99	$12.50	$24.99	ISL
See3CAM_CU135	11	e-con Systems	Sensor	See3CAM_CU135_CHL_TC_BX	1	1	$359.00	$359.00	$359.00	ISL
Severe Weather 4-in x 4-in x 10-ft #2 Southern yellow pine Ground c	12	Lowes	Stand	312624	3	1	$14.88	$14.88	$44.64	ISL
Severe Weather 2-in x 4-in x 8-ft #2 Prime Southern yellow pine Abo	13	Lowes	Stand	196087	2	1	$3.98	$3.98	$7.96	ISL
Simpson Strong-Tie - Strong-Drive SD #9 x 1-1/2-in Mechanically	14	Lowes	Stand	130014	2	100	$15.98	$0.16	$31.96	ISL
Simpson Strong-Tie LS 3.38-in x 3.841-in x 1.59-in 18 -Gauge G90 g	15	Lowes	Stand	1944269	10	1	$2.98	$2.98	$29.80	ISL
Power Pro - #9 x 2-1/2-in Yellow zinc Interior Trim screws (100 pack)	16	Lowes	Stand	1147891	1	100	$12.98	$0.13	$12.98	ISL
Simpson Strong-Tie - 7-in 18 -Gauge ZMAX Steel Tie plate	17	Lowes	Stand	312976	2	1	$2.98	$2.98	$5.96	ISL
Severe Weather 2-in x 4-in x 10-ft #2 Prime Southern yellow pine Ab	18	Lowes	Stand	196453	2	1	$7.18	$7.18	$14.36	ISL
Galvanized Steel Corner Bracket 6.5" x 6.5" x 1.25"	19	McMaster-Carr	Stand	17715A12	8	1	$10.31	$10.31	$82.48	ISL
Galvanized Steel Corner Bracket 2-5/8" x 2-5/8" x 2"	20	McMaster-Carr	Stand	17715A11	6	1	$4.71	$4.71	$28.26	ISL
18-8 Stainless Steel Washer for 7/16" Screw Size, General Purpose,	21	McMaster-Carr	Stand	92141A032	4	100	$11.29	$0.11	$11.29	ISL
High-Strength Steel Hex Nut SAE Grade 8, 1/2"-20 Thread Size	22	McMaster-Carr	Stand	90499A825	4	50	$9.49	$0.09	$9.49	ISL
18-8 Stainless Steel Hex Head Screw 1/2"-20 Thread Size, 5" Long,	23	McMaster-Carr	Stand	92800A413	4	1	$2.85	$2.85	$11.40	ISL
Mighty-Lite Caster 3-5/8" x 2-1/2" Plate, Swivel with Brake and 4" 80	24	McMaster-Carr	Stand	2835T15	4	1	$17.71	$17.71	$70.84	ISL
Hex Head Screws for Wood 18-8 Stainless Steel, 5/16" Size, 1-1/2" L	25	McMaster-Carr	Stand	92351A587	20	25	$13.17	$0.53	$13.17	ISL
Black-Oxide Alloy Steel Socket Head Screw 5/16"-18 Thread Size	26	McMaster-Carr	Mount	91251A578	8	50	$17.60	$0.35	$17.60	ISL
Zinc Yellow-Chromate Plated Grade 8 Steel SAE Washer	27	McMaster-Carr	Mount	98023A116	8	50	$9.17	$0.18	$9.17	ISL
Black-Oxide Alloy Steel Socket Head Screw 1/4"-20 Thread Size	28	McMaster-Carr	Mount	91251A538	4	25	$15.43	$0.62	$15.43	ISL
Black-Oxide Alloy Steel Socket Head Screw 1/4"-20 Thread Size	29	McMaster-Carr	Mount	91251A540	8	50	$11.38	$0.23	$11.38	ISL
Zinc Yellow-Chromate Plated Grade 8 Steel SAE Washer	30	McMaster-Carr	Mount	98023A115	12	100	$11.12	$0.11	$11.12	ISL
Alloy Steel Shoulder Screw 5/16" Shoulder Diameter	31	McMaster-Carr	Mount	91259A581	2	1	$2.10	$2.10	$4.20	ISL
Oil-Embedded 841 Bronze Flanged Sleeve Bearing	32	McMaster-Carr	Mount	6338K102	2	1	$16.37	$16.37	$32.74	ISL
SAE Grade 8 High-Strength Steel Hex Nut	33	McMaster-Carr	Mount	94895A029	2	100	$5.81	$0.06	$5.81	ISL
Black-Oxide Alloy Steel Socket Head Screw 6-32 Thread Size, 5/16"	34	McMaster-Carr	Mount	91864A019	2	10	$11.47	$0.23	$11.47	ISL
Brass Washer	35	McMaster-Carr	Mount	92916A325	4	100	$3.90	$0.04	$3.90	ISL
18-8 Stainless Steel Socket Head Screw	36	McMaster-Carr	Mount	91292A118	3	100	$9.81	$0.09	$9.81	ISL
18-8 Stainless Steel General Purpose Washer	37	McMaster-Carr	Mount	93475A230	3	100	$3.57	$0.04	$3.57	ISL
Oil-Resistant Buna-N Gasket	38	McMaster-Carr	Mount	5081K88	3	1	$1.26	$1.26	$3.78	ISL
Steel DIN 3 Rail, 5-7/8" Length	39	McMaster-Carr	Mount	8961K86	1	1	$4.47	$4.47	$4.47	ISL
Black-Oxide Alliy Steel Socket Head Screw 6-32 Thread Size, 1/2" L	40	McMaster-Carr	Mount	91251A148	2	100	$11.95	$8.37	$11.95	ISL
Low-Carbon Steel Bar 1/4" Thick, 5" Wide	41	McMaster-Carr	Mount	8910K591	1	1	$72.11	72.11	72,11	ISL
18-8 Stainless Steel Thread-Locking Socket Head Screw M2 x 0.4 m	42	McMaster-Carr	Mount	93705A802	4	25	$6.21	0.25	$6.21	ISL
18-8 Stainless Steel Washer for M1.91 Screw Size	43	McMaster-Carr	Mount	98689A244	4	50	$8.68	$0.17	$8.68	ISL
Low-Carbon Steel Sheet 10" x 10", 1/4" Thick	44	McMaster-Carr	Mount	6544K24	1	1	$60.07	$60.07	$60.07	ISL
Multipurpose 6061 Aluminum 90 Degree Angle	45	McMaster-Carr	Mount	8982K87	1	1	$5.20	$5.20	$5.20	ISL
Multipurpose 6061 Aluminum Bar	46	McMaster-Carr	Mount	8975K443	1	1	$29.08	$29.08	$29.08	ISL
Powder-Coated Zinc Adjustable Handle with 5/16"-18 Threaded Hole	47	McMaster-Carr	Mount	64835K96	2	1	$10.68	$10.68	$10.68	ISL
Black-Oxide Alloy Steel Socket Head Screw 5/16"-18 Thread Size	48	McMaster-Carr	Mount	91251A620	2	10	$10.49	$0.95	$10.49	ISL
Threaded-Stem Swivel Caster 1/2"-13 Thread with Brake and 4" Diar	49	McMaster-Carr	Mount	2834T47	2	1	$19.65	$19.65	$39.30	ISL
High-Strength Steel Nylon-Insert Locknut Grade 8, Zinc Yellow-Chror	50	McMaster-Carr	Mount	97135A210	2	25	$5.29	$0.21	$5.29	ISL
Snug-Fit Vibration-Damping Loop Clamp Aluminum with Neoprene C	51	McMaster-Carr	Mount	4711N118	1	5	$12.69	$2.54	$12.69	ISL
Hook and Loop Cable Strap 13" Overall Length	52	McMaster-Carr	Mount	6605K24	1	10	$12.97	$1.30	$12.97	ISL
UV-Resistant Spiral Sleeving 5/8" ID 10ft length Color: Black	53	McMaster-Carr	Electrical	7432K165	1	1	$18.83	$18.83	$18.83	ISL
4" x 6" x 0.25" Carbon Steel Rectangle Tube A500	54	Online Metals	Mount	mp-00063100	1	1	$4.10	$4.10	$4.10	ISL
8" x 3.75" x 0.25" x 0.41" Aluminum Channel 6061-T6-Extruded Alum	55	Online Metals	Mount	13562	1	1	$71.54	$71.54	$71.54	ISL
80/20® 4" Swivel Caster W/ 7/16-14 Stem W/ Brake	56	Global Industrial	Mount	WBB3247169	2	1	$13.25	$13.25	$26.50	ISL
Global Industrial™ 78"H Mobile Post with Caster Base - Black	57	Global Industrial	Mount	WB239200BK	1	1	$439.95	$439.95	$439.95	Supporter
Global Industrial™ Plastic Laminate Worksurface	58	Global Industrial	Mount	WB242257BK	1	1	$149.95	$149.95	$149.95	Supporter
Footrest For Global Industrial™ Orbit Computer Workstations	59	Global Industrial	Mount	WB752150BK	1	1	$39.99	$39.99	$39.99	Supporter
Global Industrial™ Power Kit for 68"H and 81"H Orbit Posts - Black	60	Global Industrial	Electrical	WB670071BK	1	1	$70.95	$70.95	$70.95	Supporter
Polarizing Filter	61	OpenMV	Sensor	OMV-POLARIZING-FILTER	1	1	$15.00	$15.00	$15.00	Ryan
Grade 2 Titanium General Purpose Washer for Number 6 Screw Size	62	McMaster-Carr	Mount	94051A208	2	5	$9.74	$1.95	$9.74	Cris
Zinc Yellow-Chromate Plated Hex Head Screw Grade 8 Steel, 1/4"-2	63	McMaster-Carr	Mount	92620A539	8	100	$20.37	$0.20	$20.37	ISL Excess
18-8 Stainless Steel Socket Head Screw M4 x 0.7 mm Thread, 20 mm	64	McMaster-Carr	Mount	91292A121	3	100	$11.89	$0.12	$11.29	Alejandro
Scotch Extreme Interlocking Fasteners, 4 Strips, 1" x 3", Delivers Pow	65	Amazon	Mount	B001LTTNOI	2	4	$4.59	$1.15	$4.59	Cris
CableCreation USB to RS232 Adapter with PL2303 Chipset, 6.6ft USB	66	Amazon	Electrical	B0769DVQM1	1	1	$11.99	$11.99	$11.99	Ryan
SanDisk 256GB ImageMate microSDXC	61	Amazon	Electrical	B08ZNTY256	1	1	$32.99	$32.99	$32.99	Ryan

Figure 9. Bill of Materials

10 Budget

As mentioned previously, the budget is considered to be the total build cost of the prototype. This is the cost of what is needed to replicate the design. The expenses for this budget were segmented into four sections: Mount, Electrical, Sensor, and Stand. The Mount includes all

Global Industrial parts, material stock and fasteners needed to manufacture the sensing head unit. The electrical category contains all electronic devices and hardware, such as the projector and Jetson Nano. The Sensor includes the components needed for the sensing unit of the door, which is the CU135 Camera and a polarizing filter add-on. Finally, the Stand is the lumber and fasteners needed to create a stand to hold up the Cascadia truck door for testing purposes. Figure 10 displays the budget overview with percentages for each category.

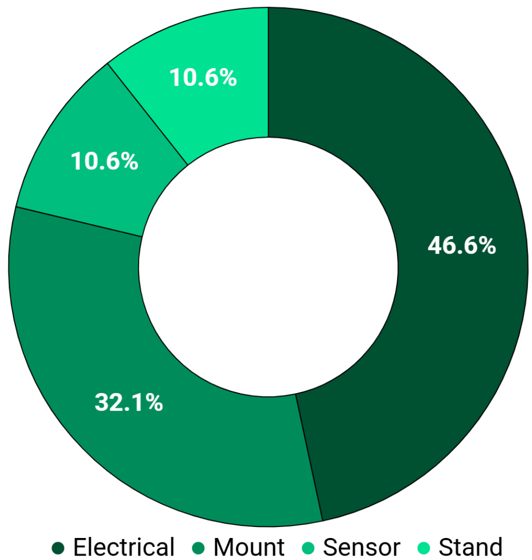


Figure 10. Budget Overview

The total build cost to replicate the automated detection system is $3,519.52 covering all previously mentioned hardware and materials. An itemized list of this build cost is provided in Figure 11.

Name	ID	Source	Subsystem	Part Code & URL	Qty Used	Quantity per Pack	Price	Unit Price	Cost	Purchaser
ViewSonic LS740HD 1080p Laser Projector	1	Amazon	Electrical	B0CDG3JKQP	1	1	$1,069.02	$1,069.02	$1,069.02	ISL
NVIDIA Jetson Orin Nano Dev Kit	2	Amazon	Electrical	B0DNZC8JGZ	1	1	$399.00	$399.00	$399.00	ISL
NVIDIA Jetson Enclosure	3	Amazon	Electrical	B0CG38BS5S	1	1	$22.99	$22.99	$22.99	ISL
Din Rail Mount Clip Snap	4	Amazon	Mount	B0B7VB7SVK	1	4	$7.99	$1.99	$7.99	ISL
Multi Purpose Rubber Spacer	5	Amazon	Mount	B0DQY4Y9B3	2	18	$6.99	$0.38	$6.99	ISL
HDMI to DisplayPort cable 6ft	6	Amazon	Electrical	B0BV2KKX2R	1	1	$6.79	$6.79	$6.79	ISL
AGVEE 2 Pack Mid Angle USB4 USB-C Male to USB-C Female Ada	7	Amazon	Electrical	B0DQXNB3DN	1	1	$8.99	$8.99	$8.99	ISL
Under Desk Cable Management- 2 Pack Cord Organizer. Metal Wire	8	Amazon	Mount	B08KFT479X	1	2	$24.99	$12.50	$24.99	ISL
See3CAM_CU135	9	e-con Systems	Sensor	See3CAM_CU135_CHL_TC_BX	1	1	$359.00	$359.00	$359.00	ISL
Severe Weather 4-in x 4-in x 10-ft #2 Southern yellow pine Ground c	10	Lowes	Stand	312624	3	1	$14.88	$14.88	$44.64	ISL
Severe Weather 2-in x 4-in x 8-ft #2 Prime Southern yellow pine Abo	11	Lowes	Stand	196087	2	1	$3.98	$3.98	$7.96	ISL
Simpson Strong-Tie - Strong-Drive SD #9 x 1-1/2-in Mechanically	12	Lowes	Stand	130014	2	100	$15.98	$0.16	$31.96	ISL
Simpson Strong-Tie LS 3.38-in x 3.841-in x 1.59-in 18 -Gauge G90 g	13	Lowes	Stand	1944269	10	1	$2.98	$2.98	$29.80	ISL
Power Pro - #9 x 2-1/2-in Yellow zinc Interior Trim screws (100 pack)	14	Lowes	Stand	1147891	1	100	$12.98	$0.13	$12.98	ISL
Simpson Strong-Tie - 7-in 16 -Gauge ZMAX Steel Tie plate	15	Lowes	Stand	312976	2	1	$2.98	$2.98	$5.96	ISL
Severe Weather 2-in x 4-in x 10-ft #2 Prime Southern yellow pine Ab	16	Lowes	Stand	196453	2	1	$7.18	$7.18	$14.36	ISL
Galvanized Steel Corner Bracket 6.5" x 6.5" x 1.25"	17	McMaster-Carr	Stand	17715A12	8	1	$10.31	$10.31	$82.48	ISL
Galvanized Steel Corner Bracket 2-5/8" x 2-5/8" x 2"	18	McMaster-Carr	Stand	17715A11	6	1	$4.71	$4.71	$28.26	ISL
18-8 Stainless Steel Washer for 7/16" Screw Size, General Purpose,	19	McMaster-Carr	Stand	92141A032	4	100	$11.29	$0.11	$11.29	ISL
High-Strength Steel Hex Nut SAE Grade 8, 1/2"-20 Thread Size	20	McMaster-Carr	Stand	90499A825	4	50	$9.49	$0.09	$9.49	ISL
18-8 Stainless Steel Hex Head Screw 1/2"-20 Thread Size, 5" Long,	21	McMaster-Carr	Stand	92800A413	4	1	$2.85	$2.85	$11.40	ISL
Mighty-Lite Caster 3-5/8" x 2-1/2" Plate, Swivel with Brake and 4" 80	22	McMaster-Carr	Stand	2835T15	4	1	$17.71	$17.71	$70.84	ISL
Hex Head Screws for Wood 18-8 Stainless Steel, 5/16" Size, 1-1/2" L	23	McMaster-Carr	Stand	92351A587	20	25	$13.17	$0.53	$13.17	ISL
Black-Oxide Alloy Steel Socket Head Screw 5/16"-18 Thread Size	24	McMaster-Carr	Mount	91251A578	8	50	$17.60	$0.35	$17.60	ISL
Zinc Yellow-Chromate Plated Grade 8 Steel SAE Washer	25	McMaster-Carr	Mount	98023A116	8	50	$9.17	$0.18	$9.17	ISL
Black-Oxide Alloy Steel Socket Head Screw 1/4"-20 Thread Size	26	McMaster-Carr	Mount	91251A538	4	25	$15.43	$0.62	$15.43	ISL
Black-Oxide Alloy Steel Socket Head Screw 1/4"-20 Thread Size	27	McMaster-Carr	Mount	91251A540	8	50	$11.38	$0.23	$11.38	ISL
Zinc Yellow-Chromate Plated Grade 8 Steel SAE Washer	28	McMaster-Carr	Mount	98023A115	12	100	$11.12	$0.11	$11.12	ISL
Alloy Steel Shoulder Screw 5/16" Shoulder Diameter	29	McMaster-Carr	Mount	91259A581	2	1	$2.10	$2.10	$4.20	ISL
Oil-Embedded 841 Bronze Flanged Sleeve Bearing	30	McMaster-Carr	Mount	6338K102	2	1	$16.37	$16.37	$32.74	ISL
Black-Oxide Alloy Steel Socket Head Screw 6-32 Thread Size, 5/16"	31	McMaster-Carr	Mount	91864A019	2	10	$11.47	$0.23	$11.47	ISL
Brass Washer	32	McMaster-Carr	Mount	92916A325	4	100	$3.90	$0.04	$3.90	ISL
Oil-Resistant Buna-N Gasket	33	McMaster-Carr	Mount	5081K88	3	1	$1.26	$1.26	$3.78	ISL
Steel DIN 3 Rail, 5-7/8" Length	34	McMaster-Carr	Mount	8961K86	1	1	$4.47	$4.47	$4.47	ISL
Black-Oxide Alliy Steel Socket Head Screw 6-32 Thread Size, 1/2" L	35	McMaster-Carr	Mount	91251A148	2	100	$11.95	$8.37	$11.95	ISL
Low-Carbon Steel Bar 1/4" Thick, 5" Wide	36	McMaster-Carr	Mount	8910K591	1	1	$72.11	72.11	72,11	ISL
18-8 Stainless Steel Thread-Locking Socket Head Screw M2 x 0.4 m	37	McMaster-Carr	Mount	93705A802	4	25	$6.21	0.25	$6.21	ISL
18-8 Stainless Steel Washer for M1.91 Screw Size	38	McMaster-Carr	Mount	98689A244	4	50	$8.68	$0.17	$8.68	ISL
Low-Carbon Steel Sheet 10" x 10", 1/4" Thick	39	McMaster-Carr	Mount	6544K24	1	1	$60.07	$60.07	$60.07	ISL
Multipurpose 6061 Aluminum 90 Degree Angle	40	McMaster-Carr	Mount	8982K87	1	1	$5.20	$5.20	$5.20	ISL
Multipurpose 6061 Aluminum Bar	41	McMaster-Carr	Mount	8975K443	1	1	$29.08	$29.08	$29.08	ISL
Powder-Coated Zinc Adjustable Handle with 5/16"-18 Threaded Hole	42	McMaster-Carr	Mount	64835K96	2	1	$10.68	$10.68	$10.68	ISL
Black-Oxide Alloy Steel Socket Head Screw 5/16"-18 Thread Size	43	McMaster-Carr	Mount	91251A620	2	10	$10.49	$0.95	$10.49	ISL
Threaded-Stem Swivel Caster 1/2"-13 Thread with Brake and 4" Dia	44	McMaster-Carr	Mount	2834T47	2	1	$19.65	$19.65	$39.30	ISL
High-Strength Steel Nylon-Insert Locknut Grade 8, Zinc Yellow-Chro	45	McMaster-Carr	Mount	97135A210	2	25	$5.29	$0.21	$5.29	ISL
Snug-Fit Vibration-Damping Loop Clamp Aluminum with Neoprene C	46	McMaster-Carr	Mount	4711N118	1	5	$12.69	$2.54	$12.69	ISL
Hook and Loop Cable Strap 13" Overall Length	47	McMaster-Carr	Mount	6605K24	1	10	$12.97	$1.30	$12.97	ISL
UV-Resistant Spiral Sleeving 5/8" ID 10ft length Color: Black	48	McMaster-Carr	Electrical	7432K165	1	1	$18.83	$18.83	$18.83	ISL
4" x 6" x 0.25" Carbon Steel Rectangle Tube A500	49	Online Metals	Mount	mp-00063100	1	1	$4.10	$4.10	$4.10	ISL
8" x 3.75" x 0.25" x 0.41" Aluminum Channel 6061-T6-Extruded Alum	50	Online Metals	Mount	13562	1	1	$71.54	$71.54	$71.54	ISL
Global Industrial™ 78"H Mobile Post with Caster Base - Black	51	Global Industrial	Mount	WB239200BK	1	1	$439.95	$439.95	$439.95	Supporter
Global Industrial™ Plastic Laminate Worksurface	52	Global Industrial	Mount	WB242257BK	1	1	$149.95	$149.95	$149.95	Supporter
Footrest For Global Industrial™ Orbit Computer Workstations	53	Global Industrial	Mount	WB752150BK	1	1	$39.99	$39.99	$39.99	Supporter
Global Industrial™ Power Kit for 68"H and 81"H Orbit Posts - Black	54	Global Industrial	Electrical	WB670071BK	1	1	$70.95	$70.95	$70.95	Supporter
Polarizing Filter	55	OpenMV	Sensor	OMV-POLARIZING-FILTER	1	1	$15.00	$15.00	$15.00	Ryan
Grade 2 Titanium General Purpose Washer for Number 6 Screw Size	56	McMaster-Carr	Mount	94051A208	2	5	$9.74	$1.95	$9.74	Cris
Zinc Yellow-Chromate Plated Hex Head Screw Grade 8 Steel, 1/4"-2	57	McMaster-Carr	Mount	92620A539	8	100	$20.37	$0.20	$20.37	ISL Excess
18-8 Stainless Steel Socket Head Screw M4 x 0.7 mm Thread, 20 mm	58	McMaster-Carr	Mount	91292A121	3	100	$11.89	$0.12	$11.29	Alejandro
Scotch Extreme Interlocking Fasteners, 4 Strips, 1" x 3", Delivers Pow	59	Amazon	Mount	B001LTTNOI	2	4	$4.59	$1.15	$4.59	Cris
CableCreation USB to RS232 Adapter with PL2303 Chipset, 6.6ft USI	60	Amazon	Electrical	B0769DVQM1	1	1	$11.99	$11.99	$11.99	Ryan
SanDisk 256GB ImageMate microSDXC	61	Amazon	Electrical	B0B7NTY2S6	1	1	$32.99	$32.99	$32.99	Ryan

Figure 11. Comprehensive Build Cost List

11 Conclusion

Fontaine Modification requested a design and prototype for a projection system that can automatically identify a truck door and display an associated decal layout. The FONTAINE_DECAL team has successfully fabricated and programmed an automated projection system. This was done through the fabrication of a specialized sensing head for all hardware mounting and a robust mobile frame for shop mobility. Through the use of segmentation mask, 3D CAD coordinate mapping, image warping and homography calculations the team was able to create a precise decal projection system. This system was validated through in-lab and onsite testing. All documentation such as engineering drawing, user manuals, and bill of materials ensures that Fontaine Modification can easily reproduce this prototype for future production. The implantation of this design will significantly reduce operator error and increase their confidence for quick and easy decal placements.

12 References

OpenCV Development Team. 2023. *OpenCV Documentation: Feature Detection, Homography, and Projection Methods.* OpenCV. https://docs.opencv.org.

OpenCV Development Team. 2023. "solvePnP — Pose Estimation." OpenCV Documentation. https://docs.opencv.org/4.x/d5/d1f/calib3d_solvePnP.html.

ViewSonic Corporation. "LS740HD Laser Projector Datasheet." Accessed September 18, 2025. https://www.viewsonic.com/global/products/sheet/LS740HD.

e-con Systems. "AR1335 4K Monochrome Camera Module Datasheet." Accessed November 3, 2025. https://www.e-consystems.com/doc-ar1335-4k-monochrome-camera.asp .

NVIDIA Corporation. "Jetson Orin Nano Developer Kit Carrier Board Specification, Version 1.3." Accessed November 3, 2025. https://developer.download.nvidia.com/assets/embedded/secure/jetson/orin_nano/docs/Jetson-Orin-Nano-DevKit-Carrier-Board-Specification_SP-11324-001_v1.3.pdf .

13 Appendices

Appendix A: CAD Drawings

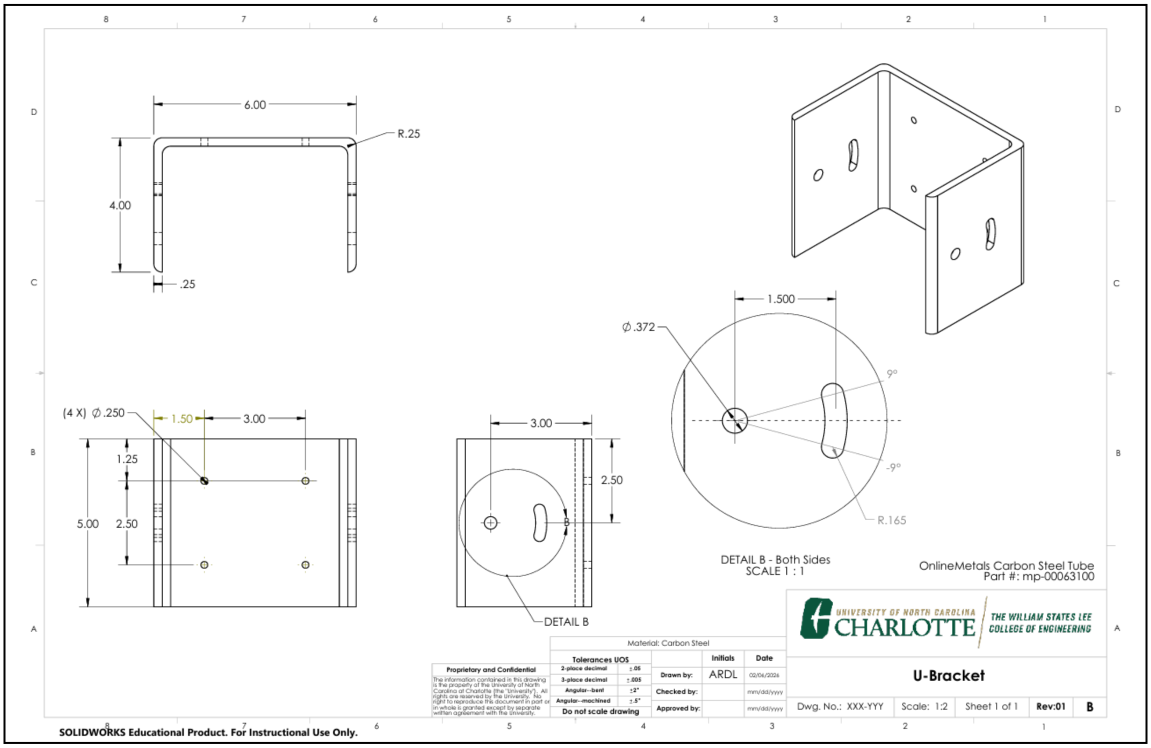


Figure 12. U-Bracket

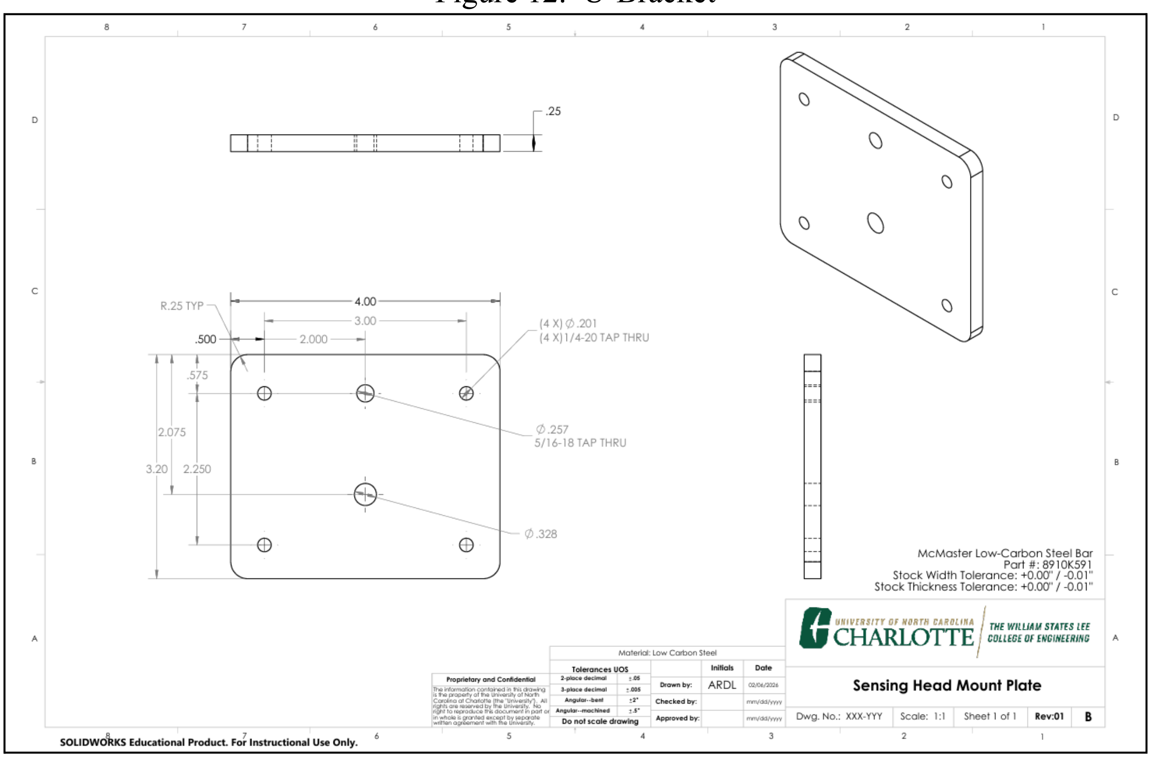


Figure 13. Sensing Head Mount Plate

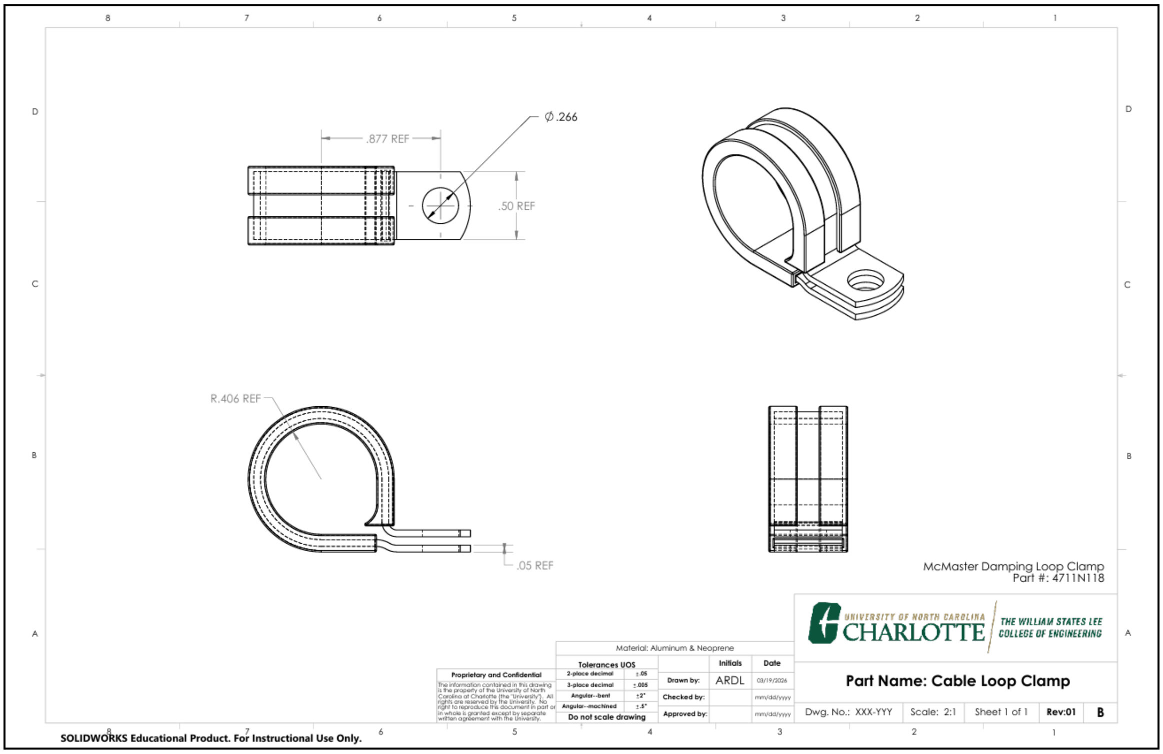


Figure 14. Cable Loop Clamp

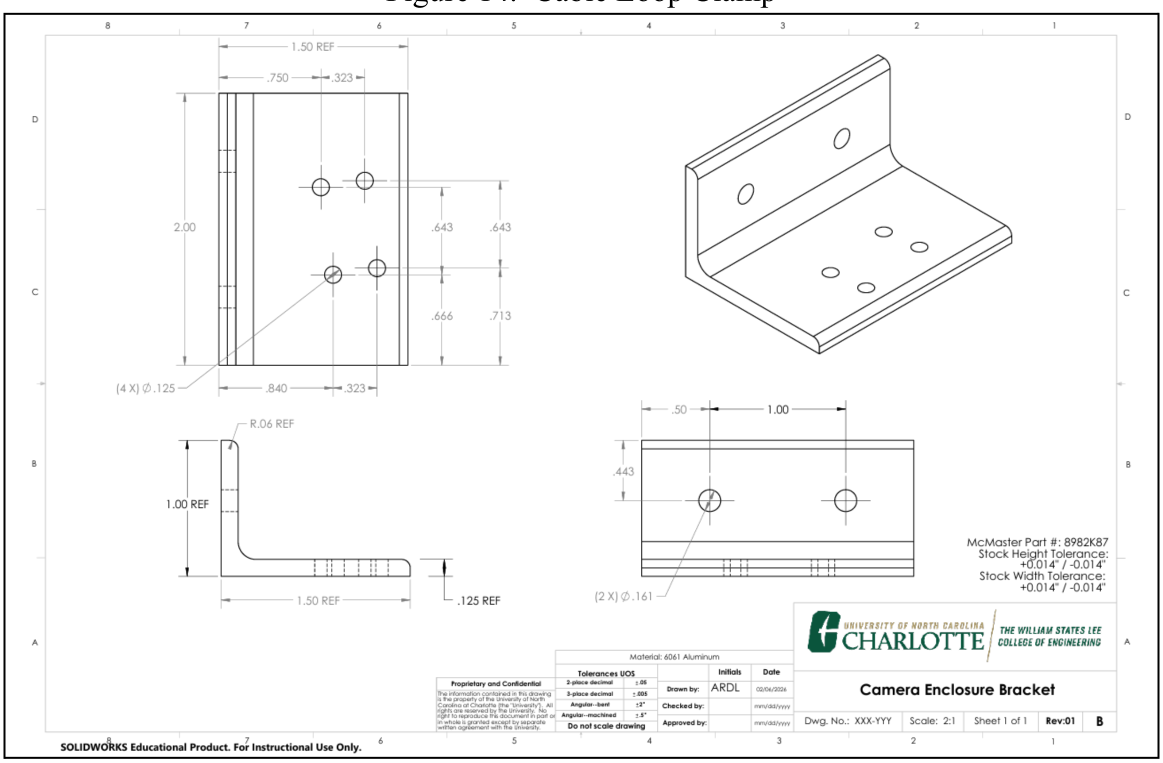


Figure 15. Camera Enclosure Bracket

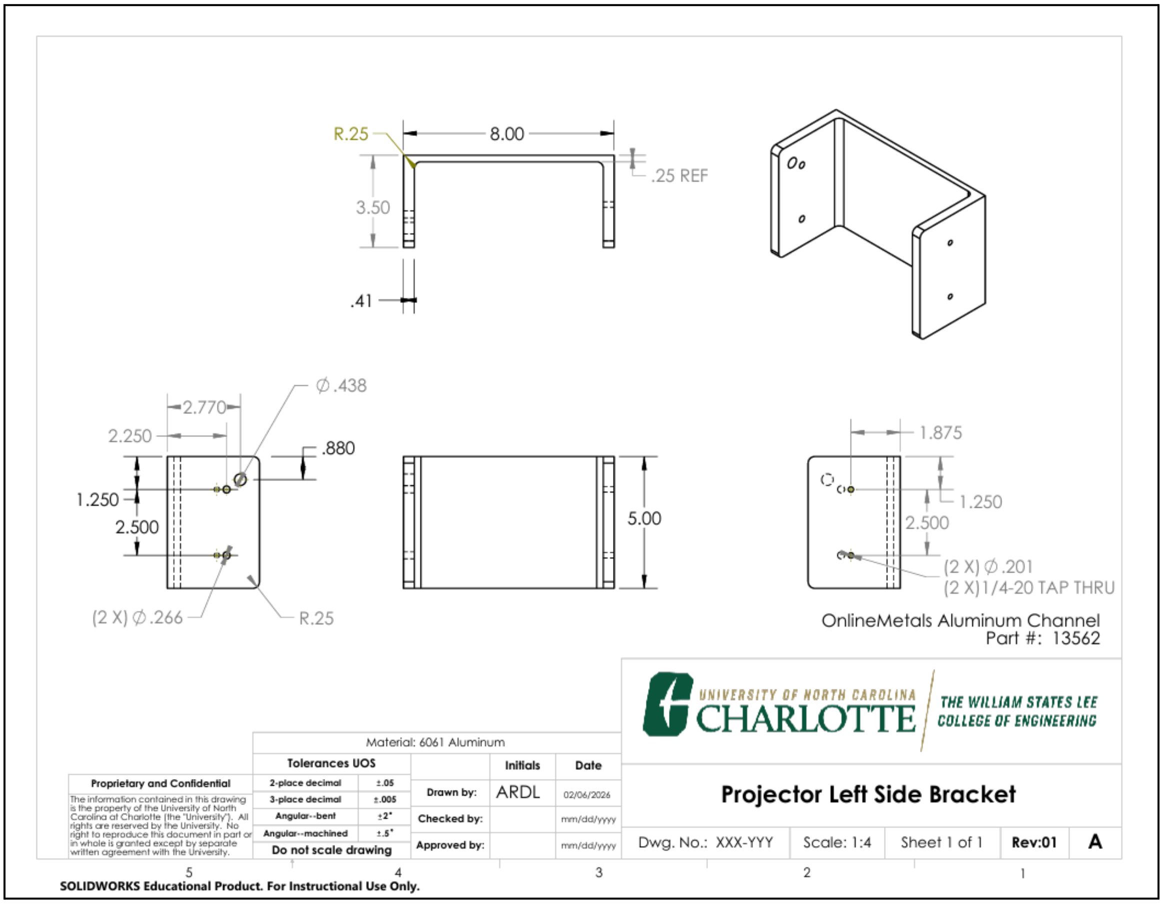


Figure 16. Projector Left Side Bracket

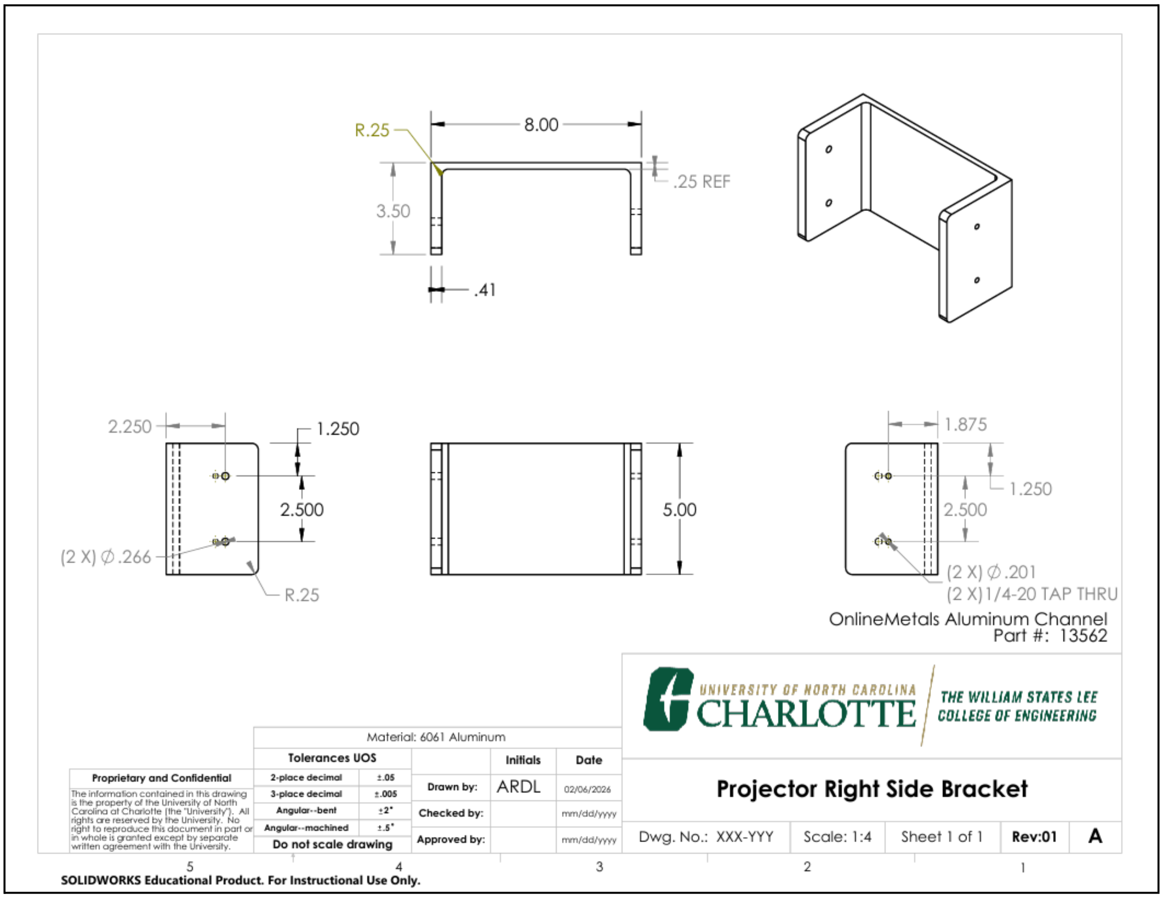


Figure 17. Projector Right Side Bracket

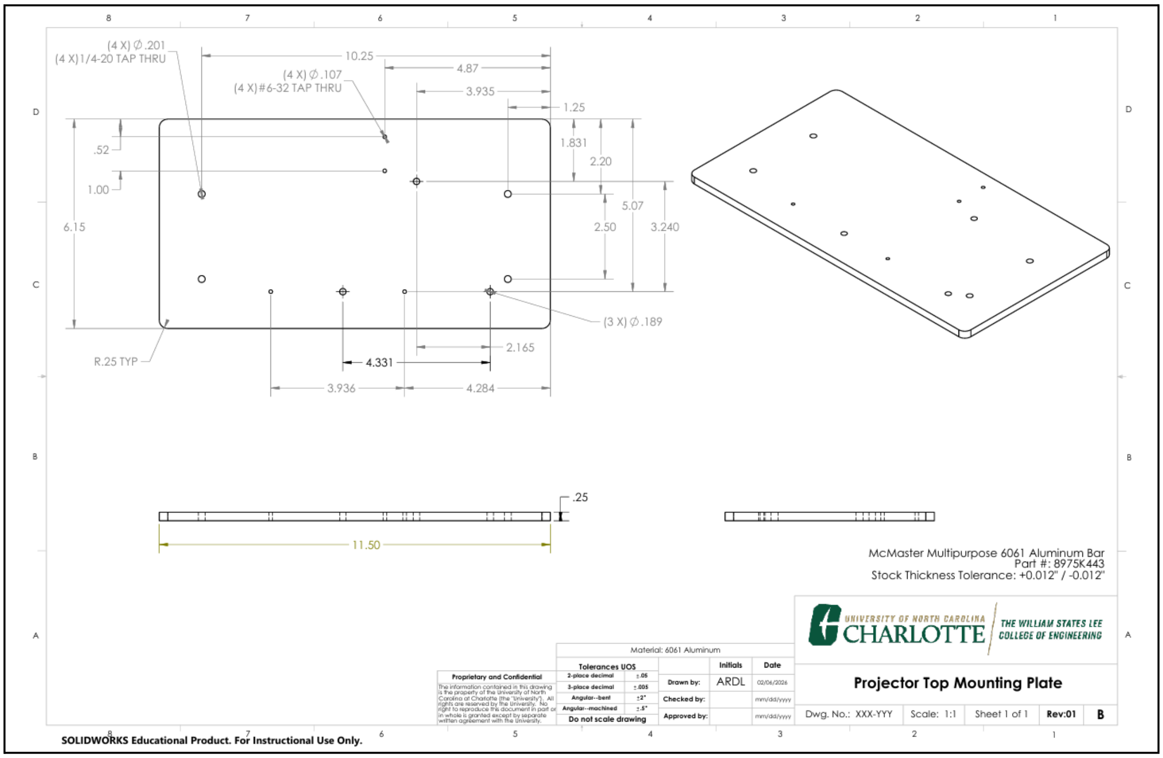


Figure 18. Projector Top Mounting Plate

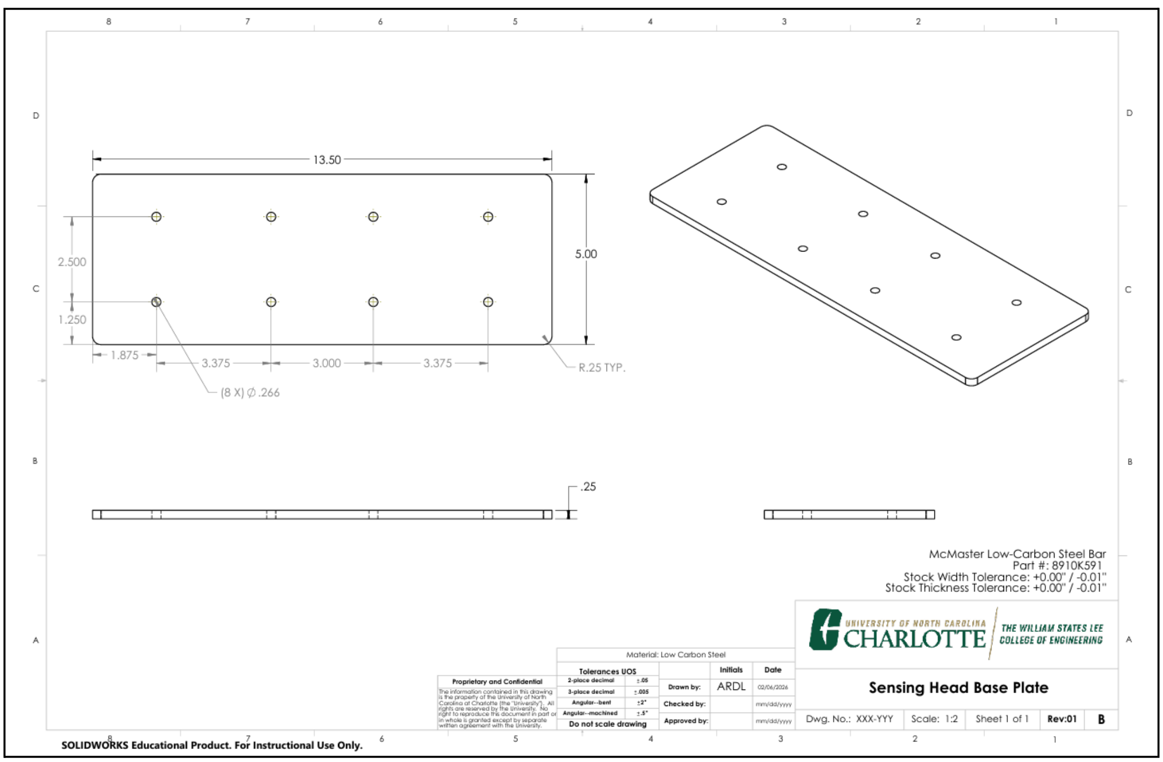
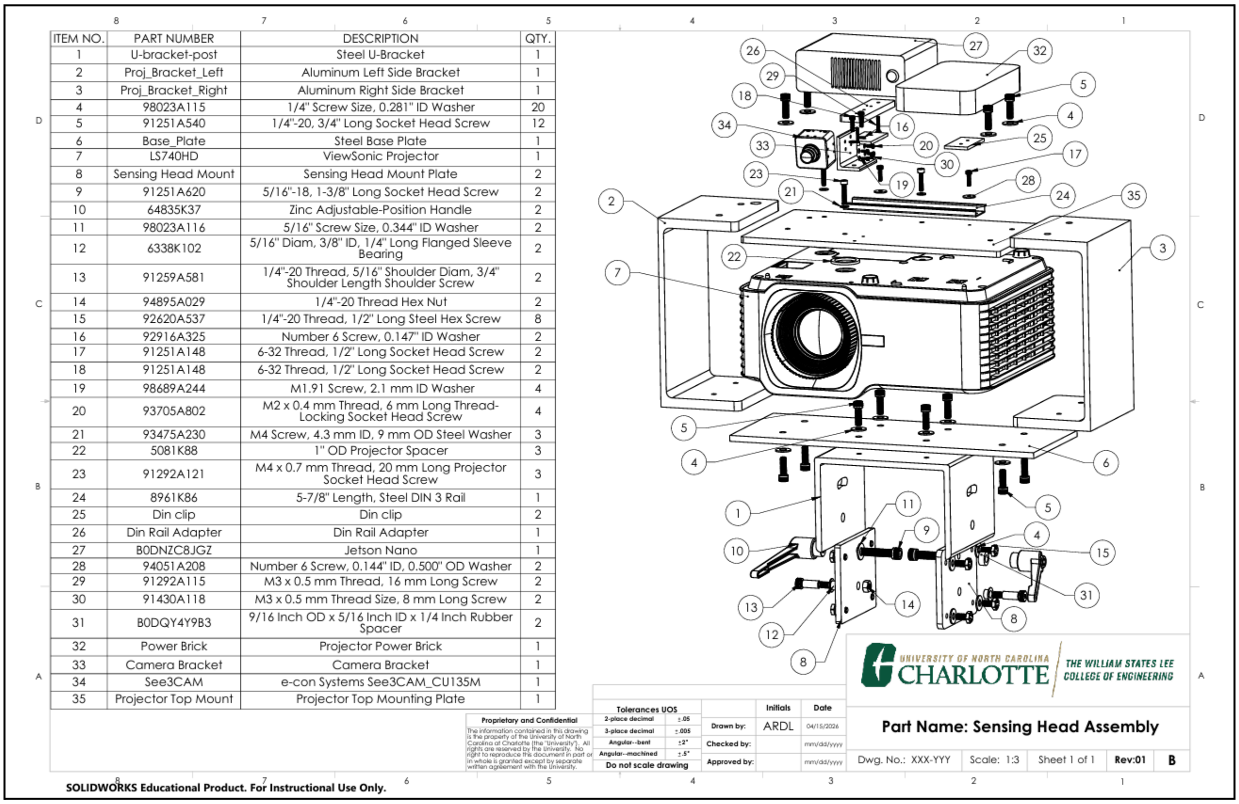


Figure 19. Sensing Head Base Plate



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	U-bracket-post	Steel U-Bracket	1
2	Proj_Bracket_Left	Aluminum Left Side Bracket	1
3	Proj_Bracket_Right	Aluminum Right Side Bracket	1
4	98023A115	1/4" Screw Size, 0.281" ID Washer	20
5	91251A540	1/4"-20, 3/4" Long Socket Head Screw	12
6	Base_Plate	Steel Base Plate	1
7	LS740HD	ViewSonic Projector	1
8	Sensing Head Mount	Sensing Head Mount Plate	2
9	91251A620	5/16"-18, 1-3/8" Long Socket Head Screw	2
10	64835K37	Zinc Adjustable-Position Handle	2
11	98023A116	5/16" Screw Size, 0.344" ID Washer	2
12	6338K102	5/16" Diam, 3/8" ID, 1/4" Long Flanged Sleeve Bearing	2
13	91259A581	1/4"-20 Thread, 5/16" Shoulder Diam, 3/4" Shoulder Length Shoulder Screw	2
14	94895A029	1/4"-20 Thread Hex Nut	2
15	92620A537	1/4"-20 Thread, 1/2" Long Steel Hex Screw	8
16	92916A325	Number 6 Screw, 0.147" ID Washer	2
17	91251A148	6-32 Thread, 1/2" Long Socket Head Screw	2
18	91251A148	6-32 Thread, 1/2" Long Socket Head Screw	2
19	98689A244	M1.91 Screw, 2.1 mm ID Washer	4
20	93705A802	M2 x 0.4 mm Thread, 6 mm Long Thread-Locking Socket Head Screw	4
21	93475A230	M4 Screw, 4.3 mm ID, 9 mm OD Steel Washer	3
22	5081K88	1" OD Projector Spacer	3
23	91292A121	M4 x 0.7 mm Thread, 20 mm Long Projector Socket Head Screw	3
24	8961K86	5-7/8" Length, Steel DIN 3 Rail	1
25	Din clip	Din clip	2
26	Din Rail Adapter	Din Rail Adapter	1
27	B0DNZC8JGZ	Jetson Nano	1
28	94051A208	Number 6 Screw, 0.144" ID, 0.500" OD Washer	2
29	91292A115	M3 x 0.5 mm Thread, 16 mm Long Screw	2
30	91430A118	M3 x 0.5 mm Thread Size, 8 mm Long Screw	2
31	B0DQY4Y9B3	9/16 Inch OD x 5/16 Inch ID x 1/4 Inch Rubber Spacer	2
32	Power Brick	Projector Power Brick	1
33	Camera Bracket	Camera Bracket	1
34	See3CAM	e-con Systems See3CAM_CU135M	1
35	Projector Top Mount	Projector Top Mounting Plate	1

Figure 20. Sensing Head Assembly

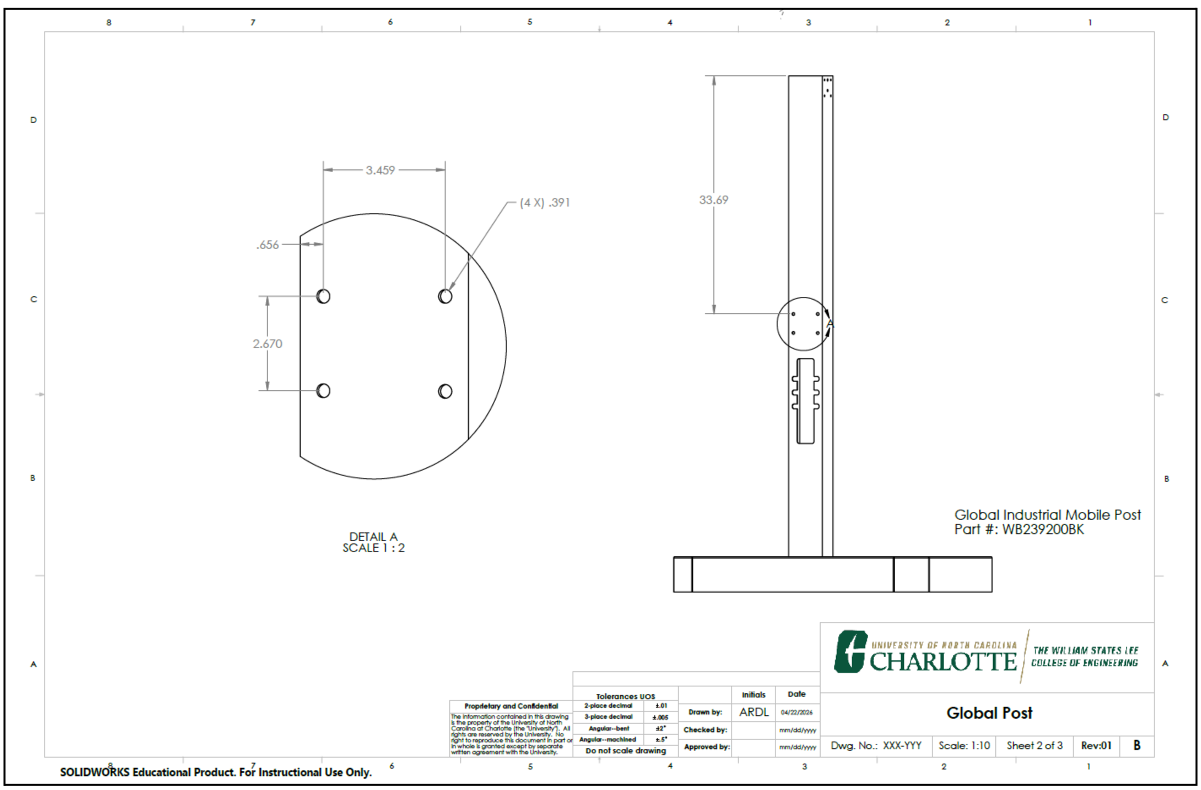


Figure 21. Global Post View A

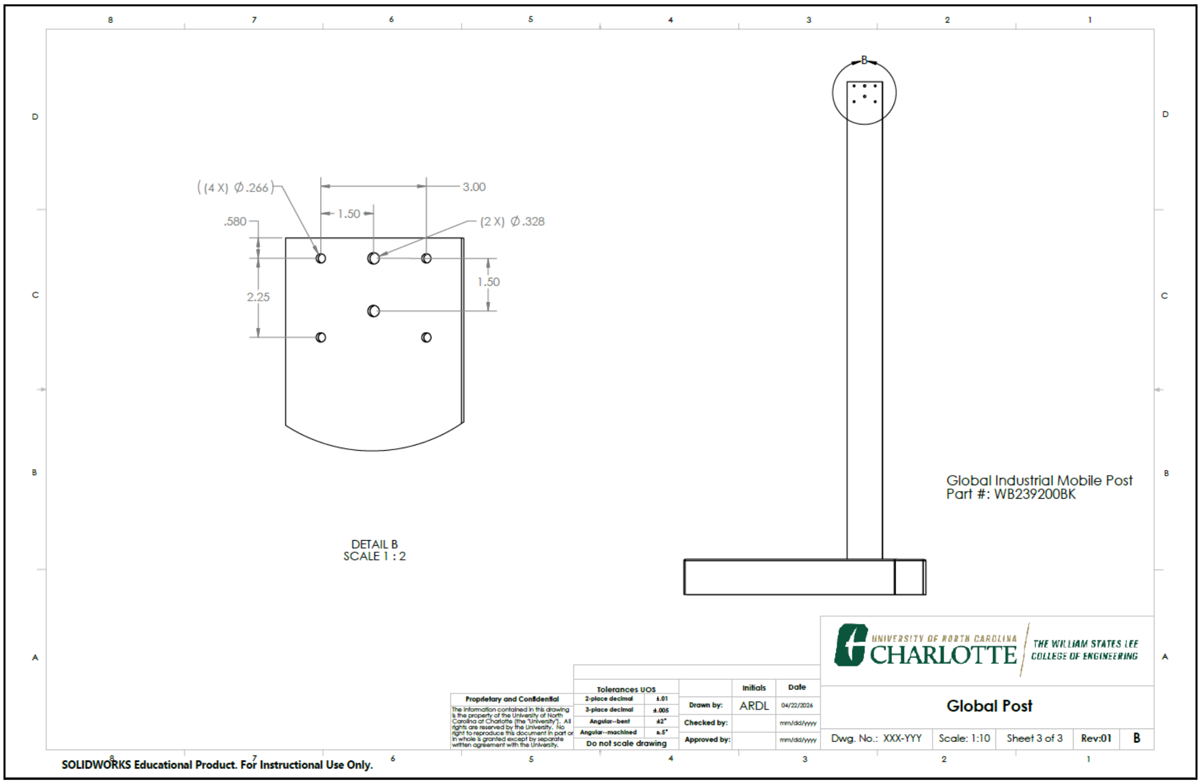
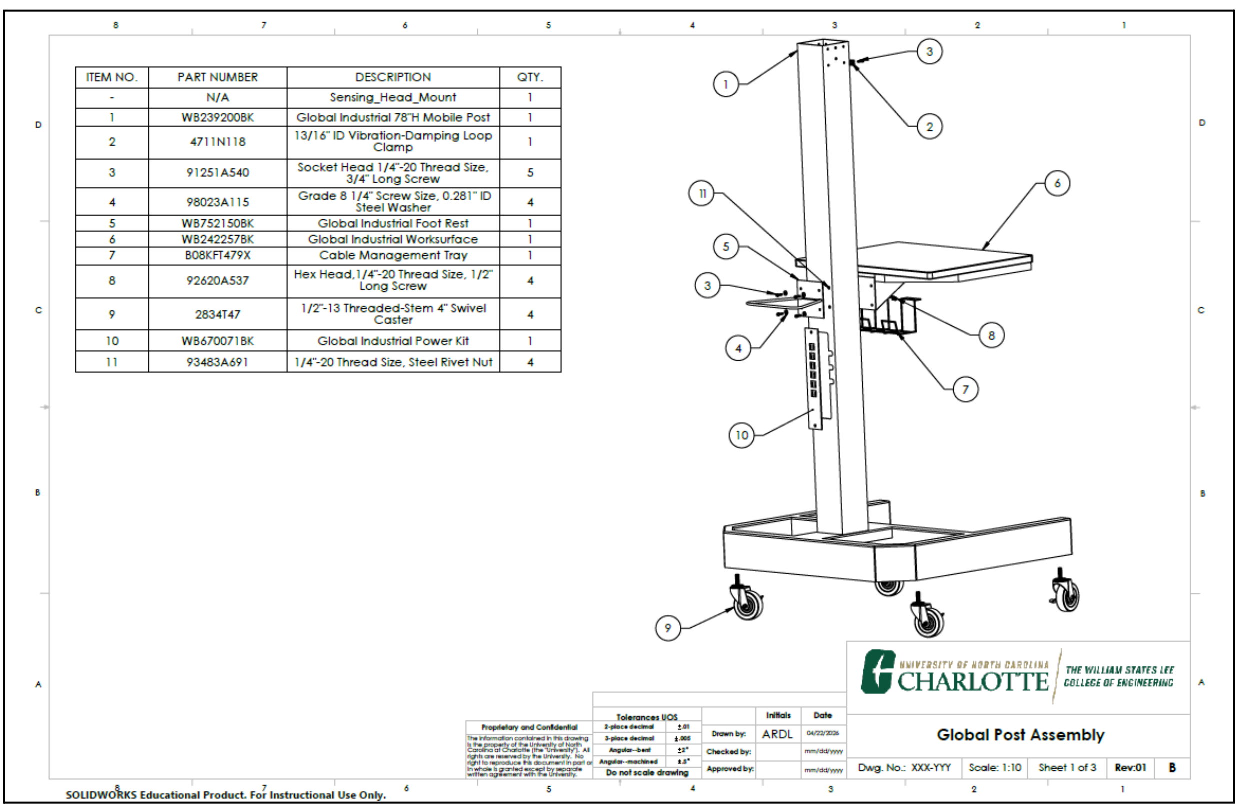


Figure 22. Global Post View B



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
-	N/A	Sensing_Head_Mount	1
1	WB239200BK	Global Industrial 78"H Mobile Post	1
2	4711N118	13/16" ID Vibration-Damping Loop Clamp	1
3	91251A540	Socket Head 1/4"-20 Thread Size, 3/4" Long Screw	5
4	98023A115	Grade 8 1/4" Screw Size, 0.281" ID Steel Washer	4
5	WB752150BK	Global Industrial Foot Rest	1
6	WB242257BK	Global Industrial Worksurface	1
7	B08KFT479X	Cable Management Tray	1
8	92620A537	Hex Head,1/4"-20 Thread Size, 1/2" Long Screw	4
9	2834T47	1/2"-13 Threaded-Stem 4" Swivel Caster	4
10	WB670071BK	Global Industrial Power Kit	1
11	93483A691	1/4"-20 Thread Size, Steel Rivet Nut	4

Figure 23. Global Post Assembly

Appendix B: Software Visual Aid

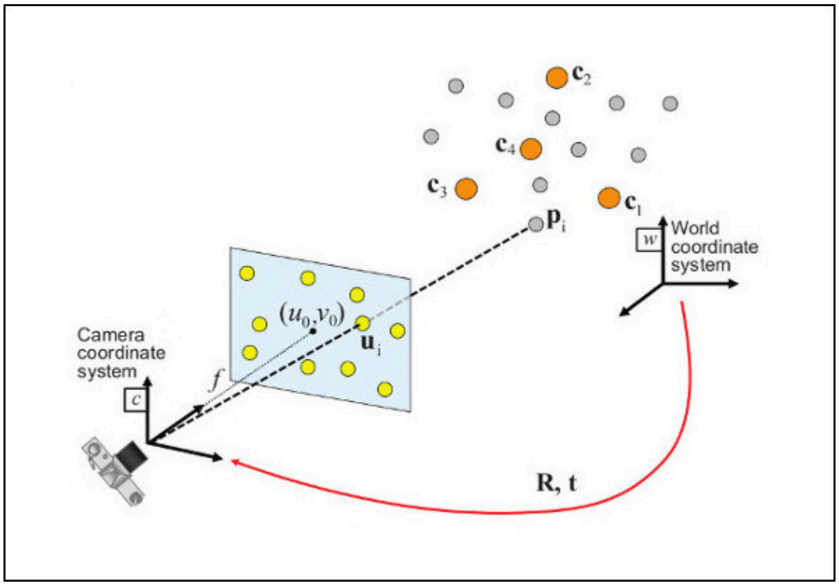


Figure 24. SolvePnP - Pose Dimension Mapping

Appendix C: Equations

```
(* Mobile Cart Tipping Calculations *)
(* All units: pounds (lb) and inches (in) *)
(* Inputs *)
Wbase = 153.61;  (* lb, base cart weight from CAD (no extra load) *)
dFbase = 23.9;   (* in, CG horizontal distance to FRONT caster line *)
dRbase = 10.1;   (* in, CG horizontal distance to REAR caster line *)
hF  = 48.0;   (* in, handle height where push/pull force is applied *)
(* Extra load on table *)
Wload = 50.0;   (* lb, load at center of table *)
dFload = 21.82;  (* in, CG -> front caster line with 50 lb *)
dRload = 12.18;  (* in, CG -> rear caster line with 50 lb *)
(* Design push/pull loads for safety factor *)
PpushDesign = 30.0;  (* lbf, expected max push *)
PpullDesign = 25.0;  (* lbf, expected max pull *)
(* Helper function: tipping force at handle *)
Ptip[W_, d_, h_] := W*d/h;
(* Case 1: Base cart only *)
PpushBase = Ptip[Wbase, dFbase, hF];  (* forward tip while pushing *)
PpullBase = Ptip[Wbase, dRbase, hF];  (* backward tip while pulling *)
SFpushBase = PpushBase/PpushDesign;
SFpullBase = PpullBase/PpullDesign;
(* Case 2: Cart + 50 lb load on table *)
Wtot = Wbase + Wload;
PpushLoad = Ptip[Wtot, dFload, hF];  (* forward tip with 50 lb *)
PpullLoad = Ptip[Wtot, dRload, hF];  (* backward tip with 50 lb *)
SFpushLoad = PpushLoad/PpushDesign;
SFpullLoad = PpullLoad/PpullDesign;
```

Out[98]=

Case	P_forward_tip (lbf)	P_backward_tip (lbf)	SF_forward (vs 30. lbf)	SF_backward (vs 25. lbf)
Base cart	76.48	32.32	2.55	1.29
Cart + 50 lb on table	92.56	51.67	3.09	2.07

Figure 25. Locked Wheel Tipping Calculations

```
In[99]:= (* Design push/pull loads for safety factor *)
PpushDesign = 30.0;  (* lbf, expected max push *)
PpullDesign = 25.0;  (* lbf, expected max pull *)

(* Helper function: tipping force at handle *)
Ptip[W_, d_, h_] := W*d/h;

(* ---- New: rolling resistance model ---- *)

Crr = 0.03;       (* rolling resistance coefficient, adjustable *)

FmoveBase = Crr*Wbase;   (* lbf, force to get cart rolling, wheels unlocked *)
FmoveLoad = Crr*Wtot;    (* lbf, with 50 lb load on table *)
```

Out[105]=

Case	P_forward_tip (lbf)	P_backward_tip (lbf)	SF_forward (vs 30. lbf)	SF_backward (vs 25. lbf)	F_move, wheels unlocked (lbf)
Base cart	76.48	32.32	2.55	1.29	4.61
Cart + 50 lb on table	92.56	51.67	3.09	2.07	6.11

Figure 26. Force To Move Cart

```
In[119]:= ClearAll["Global`*"];

(* Wheel geometry *)
R = 2.0;   (* wheel radius in inches *)
h = 0.75;   (* hose height in inches *)

(* Front wheel reactions (both front wheels combined) *)
WfrontBase = 45.63;  (* lb *)
WfrontLoad = 72.94;  (* lb *)

(* Wheel-over-step force formula *)
Fstep[W_] := W * h / Sqrt[2 * R * h - h^2];

(* Calculate forces *)
FhoseBase = Fstep[WfrontBase];
FhoseLoad = Fstep[WfrontLoad];

(* Output simple chart *)
Grid[
  {
    {"Case", "Force to Roll Over 0.75 in Hose (lbf)"},
    {"Base cart", NumberForm[FhoseBase, {6, 2}]},
    {"Cart + 50 lb on table", NumberForm[FhoseLoad, {6, 2}]}
   },
  Frame → All,
  Alignment → Center
]
```

Out[127]=

Case	Force to Roll Over 0.75 in Hose (lbf)
Base cart	21.92
Cart + 50 lb on table	35.04

Figure 27. Obstacle Climb Force Analysis (3/4in Hose)

```
# score each found line and keep best one
for entry in lines[:, 0]:
    # move local line points back to full image
    x_start, y_start, x_end, y_end = entry.astype(np.float32)
    p1 = np.array([x_start + x0, y_start + y0], dtype=np.float32)
    p2 = np.array([x_end + x0, y_end + y0], dtype=np.float32)

    # line length
    delta = p2 - p1
    length = float(np.linalg.norm(delta))
    if length < 20:
        continue

    # check if line angle fits what we want
    dx = abs(float(delta[0]))
    dy = abs(float(delta[1]))
    if orientation == "horizontal" and dx < dy * 2.0:
        continue
    if orientation == "vertical" and dy < dx * 1.3:
        continue

    # score likes long lines near expected spot
    score = length - 1.5 * float(np.linalg.norm((p1 + p2) / 2.0 - expected))
```

Figure 28. Software Calculations

Appendix D: Test Data

Figure 29. Initial Lab Decal Projection

Figure 30. Decal Projection failure as a result of segmentation error

Figure 31. Segmentation debug overlay

Figure 32. Successful Decal Overlay 4/29/26

Appendix E: Purchase Requisites

PURCHASE REQUISITION - FILL IN ALL RED ITEMS

Total: 1,530.75

Prepared Date: 11/24/25
Received date: OFFICE ONLY
Status: OFFICE ONLY — DRAFT
Mentor: Michelle Demers
Mentor email: michelle.demers@charlotte.edu
VENDOR: Amazon

ACTUAL ORDER TOTAL (ISL OFFICE)
REQUISITION /PO/TXN NO. (ISL Office)
PR: USE DROP DOWN TO IDENTIFY pr # --------> PR_1 — *Must be correct (duplicate numbers rejected)*
Member who submitted PR email : This person will receive updates via @function (function sends message directly to this email) — cburto34@charlotte.edu
Team member who submitted PR — Caleb Burton
*(Do not email purchasing for updates **before checking** with above team member or uploaded PR. Entire team has access to Team Folder/ PR/ Team Status Doc)*
Team-Save your file with team *and* PR number OR FILE WILL BE SENT BACK WITHOUT PARTS ORDERED/NO EXCEPTIONS) — Example of how to SAVE FILE: TEAM_TEST1 _1 (do not need to change here, apply to file name before sending to mentor)
DEPARTMENT — COE Senior Design
PROJECT LEAD — Caleb Burton
PROJECT LEAD EMAIL — cburto34@charlotte.edu

Justification? ->

VENDOR JUSTIFICATIONS AND QUOTES:
Include vendor name, website LINK, contact person(s) name, email address and phone number. If ordering computer equipment, please provide a reason for why this particular device is needed from this vendor. IT will need to approve all software and computers.
Justification REQUIRED WHEN:
If chosen vendor is not on the Preferred (49er Mart) Vendor list. If P-card is required (see Preferred (49er Mart) Vendor list) for vendors in RED.
QUOTE REQUIRED WHEN:
Item has specific cuts, it's stated on Preferred (49er Mart) Vendor list or it's vendor requested. If vendor is new (not on list) *check with them first. Office does not get quotes for teams.*

Justification here: "Project X needs item X from vendor X due to the special X that only vendor x can provide." or "Project X requires product X from vendor x as long lead time from other vendors will conflict with project completion."

JUSTIFICATION: Amazon is a preferred vendor. ASIN numbers are listed as "part number" below. https://www.amazon.com/ Contact: Caleb Burton, (828)707-1737, cburto34@charlotte.edu

SIGNOUT FOR PARTS: Team member who fills out PR is responsible for picking up items and noting what is missing OR coordinating with a team member before sending them to sign out for the package. Team members picking up items must make an appointment if not available to receive it during open office hours posted on Canvas and on Purchasing Status Doc. **Availability is not otherwise ensured.**

PR	VENDOR	RECEIVED DATE	SUBMITTED ORDER TOTAL	ISL PURCHASER	DATE ISL SUBMITTED	ACTUAL ORDER TOTAL	ORDER #
PR_1	Amazon		1,530.75				

ITEM NO.	QTY	PART NUMBER	PRODUCT DESCRIPTION	UNIT PRICE	TOTAL PRICE	LEAD TIMES	NOTES FOR OFFICE ONLY (DO NOT EDIT)
1	1	B0CDG3JKQP	ViewSonic LS740HD 1080p Laser Projector	1049.99	1,049.99	4 days	
2	1	B0DNZC8JGZ	NVIDIA Jetson Orin Nano Dev Kit	399.00	399.00	2 Weeks	
3	1	B0CG38BS5S	NVIDIA Jetson Enclosure	22.99	22.99	2 weeks	
4	1	B0DQY4Y9B3	uxcell 18Pcs Thick Rubber Spacer, 9/16 Inch OD x 5/16 Inch ID x 1/4 Inch Thickness Round Rubber Bushings Anti Vibration Flat Spacer Washers for Home Industrial Machinery	6.99	6.99	1 week	
5	1	B0BV2KKX2R	HDMI to DisplayPort cable 6ft	6.79	6.79	4 days	
6	1	B08PL53JG3	Cat 8 Ethernet Cable	3.99	3.99	3 days	
7	1	B0CGVB559P	25ft Extension cord with 6 outlets	21.58	21.58	3 days	
8	1	B0B7VB7SVK	4PCS 35mm Universal DIN Rail Mount Clip Snap, DIN Rail Fixed Clamp, Universal DIN Rail Mount Clip Snap in Din-Rail Mounting Brackets DIN Rail Fixed Clamp, for 35mm Din Rail, Black	7.99	7.99	1 week	
9	1	B0F8J77NYD	Canvas projector cover	11.43	11.43	3 days	

Figure 33. Purchase Requisite 1 (Amazon)

PURCHASE REQUISITION - FILL IN ALL RED ITEMS

Total: 369.60

Prepared Date: 11/24/25
Received date: OFFICE ONLY
Status: OFFICE ONLY — DRAFT
Mentor: Michelle Demers
Mentor email: michelle.demers@charlotte.edu
VENDOR: McMaster-Carr

ACTUAL ORDER TOTAL (ISL OFFICE)
REQUISITION /PO/TXN NO. (ISL Office)
PR: USE DROP DOWN TO IDENTIFY pr # --------> PR_2 — *Must be correct (duplicate numbers rejected)*
Member who submitted PR email : This person will receive updates via @function (function sends message directly to this email) — curquiz1@charlotte.edu
Team member who submitted PR — Cristian Urquiza
*(Do not email purchasing for updates **before checking** with above team member or uploaded PR. Entire team has access to Team Folder/ PR/ Team Status Doc)*
Team-Save your file with team *and* PR number OR FILE WILL BE SENT BACK WITHOUT PARTS ORDERED/NO EXCEPTIONS) — Example of how to SAVE FILE: TEAM_TEST1 _1 (do not need to change here, apply to file name before sending to mentor)
DEPARTMENT — COE Senior Design
PROJECT LEAD — Caleb Burton
PROJECT LEAD EMAIL — cburto34@charlotte.edu

Justification? ->

VENDOR JUSTIFICATIONS AND QUOTES:
Include vendor name, website LINK, contact person(s) name, email address and phone number. If ordering computer equipment, please provide a reason for why this particular device is needed from this vendor. IT will need to approve all software and computers.
Justification REQUIRED WHEN:
If chosen vendor is not on the Preferred (49er Mart) Vendor list. If P-card is required (see Preferred (49er Mart) Vendor list) for vendors in RED.
QUOTE REQUIRED WHEN:
Item has specific cuts, it's stated on Preferred (49er Mart) Vendor list or it's vendor requested. If vendor is new (not on list) *check with them first. Office does not get quotes for teams.*

Justification here: "Project X needs item X from vendor X due to the special X that only vendor x can provide." or "Project X requires product X from vendor x as long lead time from other vendors will conflict with project completion."

JUSTIFICATION: FONTAINE_DECAL requires metal stock and fastners for their cart mount. McMaster-Carr is a green preffered vendor. Cristian Urquiza, curquiz1@charlotte.edu, 336-312-0129

SIGNOUT FOR PARTS: Team member who fills out PR is responsible for picking up items and noting what is missing OR coordinating with a team member before sending them to sign out for the package. Team members picking up items must make an appointment if not available to receive it during open office hours posted on Canvas and on Purchasing Status Doc. **Availability is not otherwise ensured.**

PR	VENDOR	RECEIVED DATE	SUBMITTED ORDER TOTAL	ISL PURCHASER	DATE ISL SUBMITTED	ACTUAL ORDER TOTAL	ORDER #
PR_2	McMaster-Carr		369.60				

ITEM NO.	QTY	PART NUMBER	PRODUCT DESCRIPTION	UNIT PRICE	TOTAL PRICE	LEAD TIMES	NOTES FOR OFFICE ONLY (DO NOT EDIT)
1	1	91251A578	Black-Oxide Alloy Steel Socket Head Screw 5/16"-18 Thread Size	17.60	17.60	1 business day	
2	1	98023A116	Zinc Yellow-Chromate Plated Grade 8 Steel SAE Washer	9.17	9.17	1 business day	
3	1	91251A538	Black-Oxide Alloy Steel Socket Head Screw 1/4"-20 Thread Size	$15.43	15.43	1 business day	
4	1	91251A540	Black-Oxide Alloy Steel Socket Head Screw 1/4"-20 Thread Size	$11.38	11.38	1 business day	
5	1	98023A115	Zinc Yellow-Chromate Plated Grade 8 Steel SAE Washer with Material Certificate, for 1/4" Screw Size, 0.281" ID	$11.12	11.12	1 business day	
6	2	91259A581	Alloy Steel Shoulder Screw 5/16" Shoulder Diameter, 3/4" Shoulder Length, 1/4"-20 Thread	$2.10	4.20	1 business day	
7	2	6338K102	Oil-Embedded 841 Bronze Flanged Sleeve Bearing for 5/16" Shaft Diameter and 3/8" Housing ID, 1/4" Long	$16.37	32.74	1 business day	
8	1	94895A029	SAE Grade 8 High-Strength Steel Hex Nut Zinc-Yellow-Chromate-Plated, 1/4"-20 Thread Size	$5.81	5.81	1 business day	
9	1	91864A019	Black-Oxide Alloy Steel Socket Head Screw 6-32 Thread Size, 5/16" Long	$11.47	11.47	1 business day	
10	1	92916A325	Brass Washer for Number 6 Screw Size, General Purpose, 0.147" ID	$3.90	3.90	1 business day	
11	1	91292A118	18-8 Stainless Steel Socket Head Screw M4 x 0.7 mm Thread, 16 mm Long	$9.81	9.81	1 business day	
12	1	93475A230	18-8 Stainless Steel General Purpose Washer for M4 Screw Size, 4.3 mm ID, 9 mm OD	$3.57	3.57	1 business day	
13	3	5081K88	Oil-Resistant Buna-N Gasket for 1" Tube OD for Extra-Support Quick-Clamp Tube Fitting	$1.26	3.78	1 business day	
14	1	8961K86	Steel DIN 3 Rail, 5-7/8" Length	$4.47	4.47	1 business day	
15	1	91251A148	Black-Oxide Alloy Steel Socket Head Screw 6-32 Thread Size, 1/2" Long	$11.95	11.95	1 business day	
16	1	8910K591	Low-Carbon Steel Bar 1/4" Thick, 5" Wide	$72.11	72.11	1 business day	
17	1	93705A802	18-8 Stainless Steel Thread-Locking Socket Head Screw M2 x 0.4 mm Thread, 6 mm Long	$6.21	6.21	1 business day	
18	1	98689A244	18-8 Stainless Steel Washer for M1.91 Screw Size, General Purpose, 2.1 mm ID	$8.68	8.68	1 business day	
19	1	6544K24	Low-Carbon Steel Sheet 10" x 10", 1/4" Thick	$60.07	60.07	1 business day	
20	1	8982K87	Multipurpose 6061 Aluminum 90 Degree Angle (1 foot length)	$5.20	5.20	1 business day	
21	1	8975K443	Multipurpose 6061 Aluminum Bar (1 foot length)	$29.08	29.08	1 business day	
22	2	64835K96	Powder-Coated Zinc Adjustable Handle with 5/16"-18 Threaded Hole, Black-Oxide Insert, 2-1/8" Projection	$10.68	21.36	1 business day	
23	1	91251A620	Black-Oxide Alloy Steel Socket Head Screw 5/16"-18 Thread Size, 1-3/8" Long	$10.49	10.49	1 business day	

Figure 34. Purchase Requisite 2 (McMaster-Carr)

R21

PURCHASE REQUISITION - FILL IN ALL RED ITEMS

Prepared Date:	11/30/25	ACTUAL ORDER TOTAL (ISL OFFICE)	
Received date: OFFICE ONLY		REQUISITION /PO/TXN NO. (ISL Office)	
Status: OFFICE ONLY	DRAFT	PR: USE DROP DOWN TO IDENTIFY pr # -------->	PR_3 — *Must be correct (duplicate numbers rejected)*
		Member who submitted PR email : This person will receive updates via @function (function sends message directly to this email)	curquiz1@charlotte.edu
Mentor	Michelle Demers	Team member who submitted PR	Cristian Urquiza
Mentor email	michelle.demers@charlotte.edu	Team-Save your file with team *and* PR number OR FILE WILL BE SENT BACK WITHOUT PARTS ORDERED/NO EXCEPTIONS)	Example of how to SAVE FILE: TEAM_TEST1 _1 (do not need to change here, apply to file name before sending to mentor)
		DEPARTMENT	COE Senior Design
VENDOR	Online Metals	PROJECT LEAD	Caleb Burton
		PROJECT LEAD EMAIL	cburto34@charlotte.edu

Total 112.54

Justification? ->

*(Do not email purchasing for updates **before checking** with above team member or uploaded PR. Entire team has access to Team Folder/ PR/ Team Status Doc)*

VENDOR JUSTIFICATIONS AND QUOTES: Include vendor name, website LINK, contact person(s) name, email address and phone number. If ordering computer equipment, please provide a reason for why this particular device is needed from this vendor. IT will need to approve all software and computers.
Justification REQUIRED WHEN: If chosen vendor is not on the Preferred (49er Mart) Vendor list. If P-card is required (see Preferred (49er Mart) Vendor list) for vendors in RED.
QUOTE REQUIRED WHEN: Item has specific cuts, it's stated on Preferred (49er Mart) Vendor list or it's vendor requested. If vendor is new (not on list) *check with them first.* ***Office does not get quotes for teams.***

SIGNOUT FOR PARTS: Team member who fills out PR is responsible for picking up items and noting what is missing OR coordinating with a team member before sending them to sign out for the package. Team members picking up items must make an appointment if not available to receive it during open office hours posted on Canvas and on Purchasing Status Doc. **Availability is not otherwise ensured.**

Justification here: "Project X needs item X from vendor X due to the special X that only vendor x can provide." or "Project X requires product X from vendor x as long lead time from other vendors will conflict with project completion."

PR	VENDOR	RECEIVED DATE	SUBMITTED ORDER TOTAL	ISL PURCHASER	DATE ISL SUBMITTED	ACTUAL ORDER TOTAL	ORDER #
PR_3	Online Metals		112.54				

JUSTIFICATION: FONTAINE_DECAL requires particular metal supports for their cart mount. Online metal has specially fit metal u-channels and tubes that fit the project needs that other vendors do not supply. No green or yellow vendors have these particular fittings. https://www.onlinemetals.com. Cristian Urquiza, curquiz1@charlotte.edu, 336-312-0129

ITEM NO.	QTY	PART NUMBER	PRODUCT DESCRIPTION	UNIT PRICE	TOTAL PRICE	LEAD TIMES	NOTES FOR OFFICE ONLY (DO NOT EDIT)
1	1	mp-00063100	4" x 6" x 0.25" Carbon Steel Rectangle Tube A500 (1ft option)	41.00	41.00	2 business days	
2	1	13562	8" x 3.75" x 0.25" x 0.41" Aluminum Channel 6061-T6-Extruded Aluminum (1ft option)	71.54	71.54	2 business days	

Figure 35. Purchase Requisite 3 (Online Metals)

Z25

PURCHASE REQUISITION - FILL IN ALL RED ITEMS

Prepared Date:	12/02/25	ACTUAL ORDER TOTAL (ISL OFFICE)	
Received date: OFFICE ONLY		REQUISITION /PO/TXN NO. (ISL Office)	
Status: OFFICE ONLY	DRAFT	PR: USE DROP DOWN TO IDENTIFY pr # -------->	PR_4 — *Must be correct (duplicate numbers rejected)*
		Member who submitted PR email : This person will receive updates via @function (function sends message directly to this email)	rmonro12@charlotte.edu
Mentor	Michelle Demers	Team member who submitted PR	Ryan Monroe
Mentor email	mdemers2@charlotte.edu	Team-Save your file with team *and* PR number OR FILE WILL BE SENT BACK WITHOUT PARTS ORDERED/NO EXCEPTIONS)	Example of how to SAVE FILE: TEAM_TEST1 _1 (do not need to change here, apply to file name before sending to mentor)
		DEPARTMENT	COE Senior Design
VENDOR	e-con Systems	PROJECT LEAD	Caleb Burton
		PROJECT LEAD EMAIL	cburto34@charlotte.edu

Total 359.00

Justification? ->

*(Do not email purchasing for updates **before checking** with above team member or uploaded PR. Entire team has access to Team Folder/ PR/ Team Status Doc)*

VENDOR JUSTIFICATIONS AND QUOTES: Include vendor name, website LINK, contact person(s) name, email address and phone number. If ordering computer equipment, please provide a reason for why this particular device is needed from this vendor. IT will need to approve all software and computers.
Justification REQUIRED WHEN: If chosen vendor is not on the Preferred (49er Mart) Vendor list. If P-card is required (see Preferred (49er Mart) Vendor list) for vendors in RED.
QUOTE REQUIRED WHEN: Item has specific cuts, it's stated on Preferred (49er Mart) Vendor list or it's vendor requested. If vendor is new (not on list) *check with them first.* ***Office does not get quotes for teams.***

SIGNOUT FOR PARTS: Team member who fills out PR is responsible for picking up items and noting what is missing OR coordinating with a team member before sending them to sign out for the package. Team members picking up items must make an appointment if not available to receive it during open office hours posted on Canvas and on Purchasing Status Doc. **Availability is not otherwise ensured.**

Justification here: "Project X needs item X from vendor X due to the special X that only vendor x can provide." or "Project X requires product X from vendor x as long lead time from other vendors will conflict with project completion."

PR	VENDOR	RECEIVED DATE	SUBMITTED ORDER TOTAL	ISL PURCHASER	DATE ISL SUBMITTED	ACTUAL ORDER TOTAL	ORDER #
PR_4	e-con Systems		359.00				

JUSTIFICATION: FONTAINE_DECAL needs this high resolution camera from e-con Systems because of its compact form and narrow field-of-view that is not available in an alternative product from a preferred vendor.

ITEM NO.	QTY	PART NUMBER	PRODUCT DESCRIPTION	UNIT PRICE	TOTAL PRICE	LEAD TIMES	NOTES FOR OFFICE ONLY (DO NOT EDIT)
1	1	See3CAM_CU135_CHL_TC_BX	13MP USB Camera with Lens and Enclosure	359.00	359.00	5 business days	

Figure 36. Purchase Requisite 4 (e-con Systems)

Z25

PURCHASE REQUISITION - FILL IN ALL RED ITEMS

Prepared Date:	12/02/25	ACTUAL ORDER TOTAL (ISL OFFICE)	
Received date: OFFICE ONLY		REQUISITION /PO/TXN NO. (ISL Office)	
Status: OFFICE ONLY	DRAFT	PR: USE DROP DOWN TO IDENTIFY pr # -------->	PR_4 — *Must be correct (duplicate numbers rejected)*
		Member who submitted PR email : This person will receive updates via @function (function sends message directly to this email)	rmonro12@charlotte.edu
Mentor	Michelle Demers	Team member who submitted PR	Ryan Monroe
Mentor email	mdemers2@charlotte.edu	Team-Save your file with team *and* PR number OR FILE WILL BE SENT BACK WITHOUT PARTS ORDERED/NO EXCEPTIONS)	Example of how to SAVE FILE: TEAM_TEST1 _1 (do not need to change here, apply to file name before sending to mentor)
		DEPARTMENT	COE Senior Design
VENDOR	e-con Systems	PROJECT LEAD	Caleb Burton
		PROJECT LEAD EMAIL	cburto34@charlotte.edu

Total 359.00

Justification? ->

*(Do not email purchasing for updates **before checking** with above team member or uploaded PR. Entire team has access to Team Folder/ PR/ Team Status Doc)*

VENDOR JUSTIFICATIONS AND QUOTES: Include vendor name, website LINK, contact person(s) name, email address and phone number. If ordering computer equipment, please provide a reason for why this particular device is needed from this vendor. IT will need to approve all software and computers.
Justification REQUIRED WHEN: If chosen vendor is not on the Preferred (49er Mart) Vendor list. If P-card is required (see Preferred (49er Mart) Vendor list) for vendors in RED.
QUOTE REQUIRED WHEN: Item has specific cuts, it's stated on Preferred (49er Mart) Vendor list or it's vendor requested. If vendor is new (not on list) *check with them first.* ***Office does not get quotes for teams.***

SIGNOUT FOR PARTS: Team member who fills out PR is responsible for picking up items and noting what is missing OR coordinating with a team member before sending them to sign out for the package. Team members picking up items must make an appointment if not available to receive it during open office hours posted on Canvas and on Purchasing Status Doc. **Availability is not otherwise ensured.**

Justification here: "Project X needs item X from vendor X due to the special X that only vendor x can provide." or "Project X requires product X from vendor x as long lead time from other vendors will conflict with project completion."

PR	VENDOR	RECEIVED DATE	SUBMITTED ORDER TOTAL	ISL PURCHASER	DATE ISL SUBMITTED	ACTUAL ORDER TOTAL	ORDER #
PR_4	e-con Systems		359.00				

JUSTIFICATION: FONTAINE_DECAL needs this high resolution camera from e-con Systems because of its compact form and narrow field-of-view that is not available in an alternative product from a preferred vendor.

ITEM NO.	QTY	PART NUMBER	PRODUCT DESCRIPTION	UNIT PRICE	TOTAL PRICE	LEAD TIMES	NOTES FOR OFFICE ONLY (DO NOT EDIT)
1	1	See3CAM_CU135_CHL_TC_BX	13MP USB Camera with Lens and Enclosure	359.00	359.00	5 business days	

Figure 37. Purchase Requisite 5 (Lowes)

PURCHASE REQUISITION - FILL IN ALL RED ITEMS

Total 226.93

Prepared Date:	01/30/26	ACTUAL ORDER TOTAL (ISL OFFICE)	Justification? ->
Received date: OFFICE ONLY		REQUISITION /PO/TXN NO. (ISL Office)	
Status: OFFICE ONLY	DRAFT	PR: USE DROP DOWN TO IDENTIFY pr # -------->	PR_6 — Must be correct (duplicate numbers rejected)
		Member who submitted PR email : This person will receive updates via @function (function sends message directly to this email)	cmccoma2@charlotte.edu
Mentor	Michelle Demers	Team member who submitted PR	colton mccomas
Mentor email	mdemers2@charlotte.edu	Team-Save your file with team *and* PR number OR FILE WILL BE SENT BACK WITHOUT PARTS ORDERED/NO EXCEPTIONS)	Example of how to SAVE FILE: TEAM_TEST1 _1 (do not need to change here, apply to file name before sending to mentor)
		DEPARTMENT	COE Senior Design
VENDOR	McMaster-CARR	PROJECT LEAD	Caleb Burton
		PROJECT LEAD EMAIL	cburto34@charlotte.edu

*(Do not email purchasing for updates **before checking** with above team member or uploaded PR. Entire team has access to Team Folder/ PR/ Team Status Doc)*

PR	VENDOR	RECEIVED DATE	SUBMITTED ORDER TOTAL	ISL PURCHASER	DATE ISL SUBMITTED	ACTUAL ORDER TOTAL	ORDER #
PR_6	McMaster-CARR		226.93				

VENDOR JUSTIFICATIONS AND QUOTES: Include vendor name, website LINK, contact person(s) name, email address and phone number. If ordering computer equipment, please provide a reason for why this particular device is needed from this vendor. IT will need to approve all software and computers.
Justification REQUIRED WHEN: If chosen vendor is not on the Preferred (49er Mart) Vendor list. If P-card is required (see Preferred (49er Mart) Vendor list) for vendors in RED.
QUOTE REQUIRED WHEN: Item has specific cuts, it's stated on Preferred (49er Mart) Vendor list or it's vendor requested. If vendor is new (not on list) *check with them first.* ***Office does not get quotes for teams***.

Justification here: "Project X needs item X from vendor X due to the special X that only vendor x can provide." or "Project X requires product X from vendor x as long lead time from other vendors will conflict with project completion."

JUSTIFICATION: FONTAINE_DECAL requires a custom door-stand. McMaster-CARR will provide mounting matrerial s needed to create this tall door stand for porject prototyping. https://www.mcmaster.com/ . Colton McComas, Cmccoma2@Charlotte.edu, (919) 820-4054.

SIGNOUT FOR PARTS: Team member who fills out PR is responsible for picking up items and noting what is missing OR coordinating with a team member before sending them to sign out for the package. Team members picking up items must make an appointment if not available to receive it during open office hours posted on Canvas and on Purchasing Status Doc. **Availability is not otherwise ensured.**

ITEM NO.	QTY	PART NUMBER	PRODUCT DESCRIPTION	UNIT PRICE	TOTAL PRICE	LEAD TIMES	NOTES FOR OFFICE ONLY (DO NOT EDIT)
1	8	17715A12	Galvanized Steel Corner Bracket 6.5" x 6.5" x 1.25"	10.31	82.48	2	
2	6	17715A11	Galvanized Steel Corner Bracket 2-5/8" x 2-5/8" x 2"	4.71	28.26	2	
3	1	92141A032	18-8 Stainless Steel Washer (Pack of 100) for 7/16" Screw Size, General Purpose, 0.50" ID, 1.125" OD	11.29	11.29	2	
4	1	90499A825	High-Strength Steel Hex Nut (pack of 50) SAE Grade 8, 1/2"-20 Thread Size	9.49	9.49	2	
5	4	92800A413	18-8 Stainless Steel Hex Head Screw 1/2"-20 Thread Size, 5" Long, Fully Threaded	2.85	11.40	2	
6	4	2835T15	Mighty-Lite Caster 3-5/8" x 2-1/2" Plate, Swivel with Brake and 4" 80A Rubber Wheel	17.71	70.84		
7	1	92351A587	Hex Head Screws for Wood 18-8 Stainless Steel, 5/16" Size, 1-1/2" Long	13.17	13.17		

Figure 38. Purchase Requisite 6 (McMaster-Carr)

PURCHASE REQUISITION - FILL IN ALL RED ITEMS

Total 26.50

Prepared Date:	02/01/26	ACTUAL ORDER TOTAL (ISL OFFICE)	Justification? ->
Received date: OFFICE ONLY		REQUISITION /PO/TXN NO. (ISL Office)	
Status: OFFICE ONLY	DRAFT	PR: USE DROP DOWN TO IDENTIFY pr # -------->	PR_7 — Must be correct (duplicate numbers rejected)
		Member who submitted PR email : This person will receive updates via @function (function sends message directly to this email)	curquiz1@charlotte.edu
Mentor	Michelle Demers	Team member who submitted PR	Cristian Urquiza
Mentor email	michelle.demers@charlotte.edu	Team-Save your file with team *and* PR number OR FILE WILL BE SENT BACK WITHOUT PARTS ORDERED/NO EXCEPTIONS)	Example of how to SAVE FILE: TEAM_TEST1 _1 (do not need to change here, apply to file name before sending to mentor)
		DEPARTMENT	COE Senior Design
VENDOR	Global Industrial Equipment	PROJECT LEAD	Caleb Burton
		PROJECT LEAD EMAIL	cburto34@charlotte.edu

*(Do not email purchasing for updates **before checking** with above team member or uploaded PR. Entire team has access to Team Folder/ PR/ Team Status Doc)*

PR	VENDOR	RECEIVED DATE	SUBMITTED ORDER TOTAL	ISL PURCHASER	DATE ISL SUBMITTED	ACTUAL ORDER TOTAL	ORDER #
PR_7	Global Industrial Equipment		26.50				

VENDOR JUSTIFICATIONS AND QUOTES: Include vendor name, website LINK, contact person(s) name, email address and phone number. If ordering computer equipment, please provide a reason for why this particular device is needed from this vendor. IT will need to approve all software and computers.
Justification REQUIRED WHEN: If chosen vendor is not on the Preferred (49er Mart) Vendor list. If P-card is required (see Preferred (49er Mart) Vendor list) for vendors in RED.
QUOTE REQUIRED WHEN: Item has specific cuts, it's stated on Preferred (49er Mart) Vendor list or it's vendor requested. If vendor is new (not on list) *check with them first.* ***Office does not get quotes for teams***.

Justification here: "Project X needs item X from vendor X due to the special X that only vendor x can provide." or "Project X requires product X from vendor x as long lead time from other vendors will conflict with project completion."

JUSTIFICATION: Global Industrial Equipment is a prefered vendor. http://globalindustiral.com/ Contact: Cistian Urquiza, curquiz1@charlotte.edu. 336-312-0129

SIGNOUT FOR PARTS: Team member who fills out PR is responsible for picking up items and noting what is missing OR coordinating with a team member before sending them to sign out for the package. Team members picking up items must make an appointment if not available to receive it during open office hours posted on Canvas and on Purchasing Status Doc. **Availability is not otherwise ensured.**

ITEM NO.	QTY	PART NUMBER	PRODUCT DESCRIPTION	UNIT PRICE	TOTAL PRICE	LEAD TIMES	NOTES FOR OFFICE ONLY (DO NOT EDIT)
1	2	WBB3247169	80/20® 4" Swivel Caster W/ 7/16-14 Stem W/ Brake	13.25	26.50	8 days	

Figure 39. Purchase Requisite 7 (Global Industrial)

PURCHASE REQUISITION - FILL IN ALL RED ITEMS

Total 89.08

Prepared Date:	03/19/26	ACTUAL ORDER TOTAL (ISL OFFICE)	Justification? ->
Received date: OFFICE ONLY		REQUISITION /PO/TXN NO. (ISL Office)	
Status: OFFICE ONLY	DRAFT	PR: USE DROP DOWN TO IDENTIFY pr # -------->	PR_8 — Must be correct (duplicate numbers rejected)
		Member who submitted PR email : This person will receive updates via @function (function sends message directly to this email)	curquiz1@charlotte.edu
Mentor	Michelle Demers	Team member who submitted PR	Cristian Urquiza
Mentor email	michelle.demers@charlotte.edu	Team-Save your file with team *and* PR number OR FILE WILL BE SENT BACK WITHOUT PARTS ORDERED/NO EXCEPTIONS)	Example of how to SAVE FILE: TEAM_TEST1 _1 (do not need to change here, apply to file name before sending to mentor)
		DEPARTMENT	COE Senior Design
VENDOR	McMaster-Carr	PROJECT LEAD	Caleb Burton
		PROJECT LEAD EMAIL	cburto34@charlotte.edu

*(Do not email purchasing for updates **before checking** with above team member or uploaded PR. Entire team has access to Team Folder/ PR/ Team Status Doc)*

PR	VENDOR	RECEIVED DATE	SUBMITTED ORDER TOTAL	ISL PURCHASER	DATE ISL SUBMITTED	ACTUAL ORDER TOTAL	ORDER #
PR_8	McMaster-Carr		89.08				

VENDOR JUSTIFICATIONS AND QUOTES: Include vendor name, website LINK, contact person(s) name, email address and phone number. If ordering computer equipment, please provide a reason for why this particular device is needed from this vendor. IT will need to approve all software and computers.
Justification REQUIRED WHEN: If chosen vendor is not on the Preferred (49er Mart) Vendor list. If P-card is required (see Preferred (49er Mart) Vendor list) for vendors in RED.
QUOTE REQUIRED WHEN: Item has specific cuts, it's stated on Preferred (49er Mart) Vendor list or it's vendor requested. If vendor is new (not on list) *check with them first.* ***Office does not get quotes for teams***.

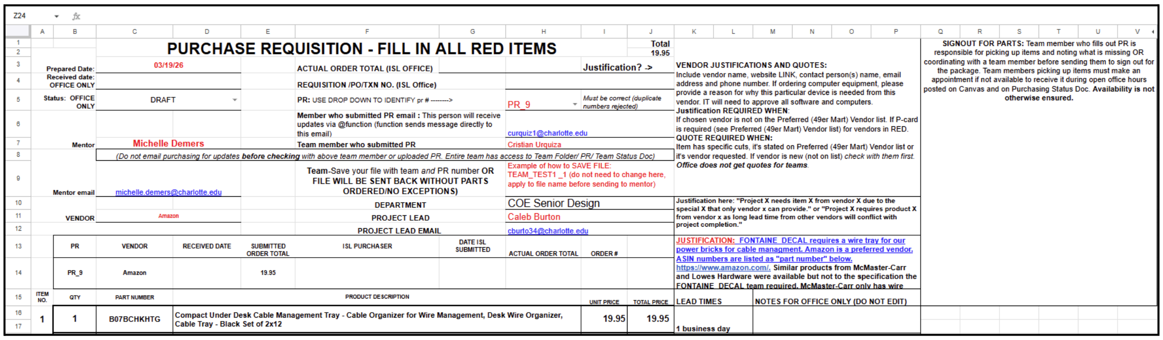
Justification here: "Project X needs item X from vendor X due to the special X that only vendor x can provide." or "Project X requires product X from vendor x as long lead time from other vendors will conflict with project completion."

JUSTIFICATION: FONTAINE_DECAL requires cable ties, sleeving and a loop clamp for cable managmnet. They alos need locking nuts and locking casters for their cart mount. McMaster-Carr is a green preffered vendor. Cristian Urquiza, curquiz1@charlotte.edu, 336-312-0129

SIGNOUT FOR PARTS: Team member who fills out PR is responsible for picking up items and noting what is missing OR coordinating with a team member before sending them to sign out for the package. Team members picking up items must make an appointment if not available to receive it during open office hours posted on Canvas and on Purchasing Status Doc. **Availability is not otherwise ensured.**

ITEM NO.	QTY	PART NUMBER	PRODUCT DESCRIPTION	UNIT PRICE	TOTAL PRICE	LEAD TIMES	NOTES FOR OFFICE ONLY (DO NOT EDIT)
1	1	97135A210	High-Strength Steel Nylon-Insert Locknut Grade 8, Zinc Yellow-Chromate Plated, 1/4"-20 Thread Size	5.29	5.29	1 business day	
2	1	4711N118	Snug-Fit Vibration-Damping Loop Clamp Aluminum with Neoprene Cushion, 13/16" ID	12.69	12.69	1 business day	
3	1	6605K24	Hook and Loop Cable Strap 13" Overall Length (Color: Black)	$12.97	12.97	1 business day	
4	2	2834T47	Threaded-Stem Swivel Caster 1/2"-13 Thread with Brake and 4" Diameter Hard Rubber Wheel	$19.65	39.30	1 business day	
5	1	7432K165	UV-Resistant Spiral Sleeving 5/8" ID (10 foot length) (Color: Black)	$18.83	18.83	1 business day	

Figure 40. Purchase Requisite 8 (McMaster-Carr)



PURCHASE REQUISITION - FILL IN ALL RED ITEMS

Total 19.95

Prepared Date: 03/19/26

Received date: OFFICE ONLY

Status: OFFICE ONLY — DRAFT

ACTUAL ORDER TOTAL (ISL OFFICE)

REQUISITION /PO/TXN NO. (ISL Office)

PR: USE DROP DOWN TO IDENTIFY pr # --------> PR_9 — Must be correct (duplicate numbers rejected)

Justification? ->

Member who submitted PR email : This person will receive updates via @function (function sends message directly to this email) — curquiz1@charlotte.edu

Mentor: Michelle Demers

Team member who submitted PR: Cristian Urquiza

(Do not email purchasing for updates **before checking** with above team member or uploaded PR. Entire team has access to Team Folder/ PR/ Team Status Doc)

Team-Save your file with team *and* PR number OR FILE WILL BE SENT BACK WITHOUT PARTS ORDERED/NO EXCEPTIONS)

Example of how to SAVE FILE: TEAM_TEST1 _1 (do not need to change here, apply to file name before sending to mentor)

Mentor email: michelle.demers@charlotte.edu

DEPARTMENT: COE Senior Design

VENDOR: Amazon

PROJECT LEAD: Caleb Burton

PROJECT LEAD EMAIL: cburto34@charlotte.edu

PR	VENDOR	RECEIVED DATE	SUBMITTED ORDER TOTAL	ISL PURCHASER	DATE ISL SUBMITTED	ACTUAL ORDER TOTAL	ORDER #
PR_9	Amazon		19.95				

ITEM NO.	QTY	PART NUMBER	PRODUCT DESCRIPTION	UNIT PRICE	TOTAL PRICE	LEAD TIMES	NOTES FOR OFFICE ONLY (DO NOT EDIT)
1	1	B07BCHKHTG	Compact Under Desk Cable Management Tray - Cable Organizer for Wire Management, Desk Wire Organizer, Cable Tray - Black Set of 2x12	19.95	19.95	1 business day	

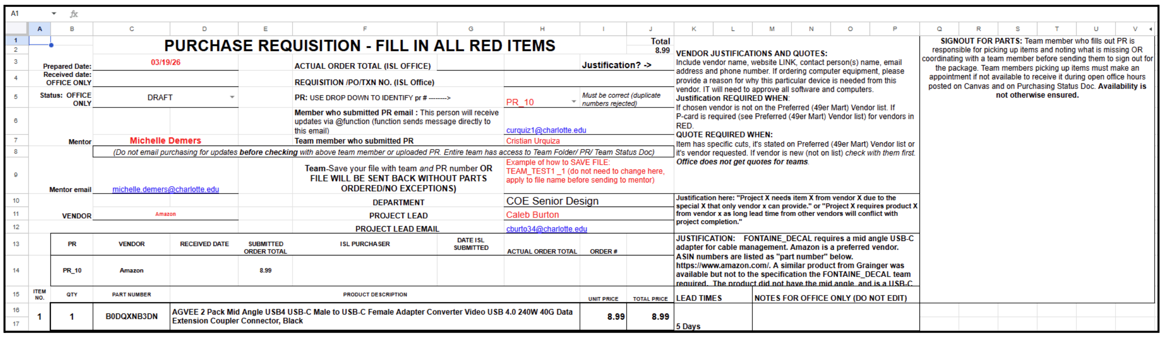
VENDOR JUSTIFICATIONS AND QUOTES:
Include vendor name, website LINK, contact person(s) name, email address and phone number. If ordering computer equipment, please provide a reason for why this particular device is needed from this vendor. IT will need to approve all software and computers.
Justification REQUIRED WHEN:
If chosen vendor is not on the Preferred (49er Mart) Vendor list. If P-card is required (see Preferred (49er Mart) Vendor list) for vendors in RED.
QUOTE REQUIRED WHEN:
Item has specific cuts, it's stated on Preferred (49er Mart) Vendor list or it's vendor requested. If vendor is new (not on list) *check with them first*.
Office does not get quotes for teams.

Justification here: "Project X needs item X from vendor X due to the special X that only vendor x can provide." or "Project X requires product X from vendor x as long lead time from other vendors will conflict with project completion."

JUSTIFICATION: FONTAINE_DECAL requires a wire tray for our power bricks for cable managment. Amazon is a preferred vendor. ASIN numbers are listed as "part number" below. https://www.amazon.com/. Similar products from McMaster-Carr and Lowes Hardware were available but not to the specification the FONTAINE_DECAL team required. McMaster-Carr only has wire

SIGNOUT FOR PARTS: Team member who fills out PR is responsible for picking up items and noting what is missing OR coordinating with a team member before sending them to sign out for the package. Team members picking up items must make an appointment if not available to receive it during open office hours posted on Canvas and on Purchasing Status Doc. **Availability is not otherwise ensured.**

Figure 41. Purchase Requisite 9 (Amazon)



PURCHASE REQUISITION - FILL IN ALL RED ITEMS

Total 8.99

Prepared Date: 03/19/26

Received date: OFFICE ONLY

Status: OFFICE ONLY — DRAFT

ACTUAL ORDER TOTAL (ISL OFFICE)

REQUISITION /PO/TXN NO. (ISL Office)

PR: USE DROP DOWN TO IDENTIFY pr # --------> PR_10 — Must be correct (duplicate numbers rejected)

Justification? ->

Member who submitted PR email : This person will receive updates via @function (function sends message directly to this email) — curquiz1@charlotte.edu

Mentor: Michelle Demers

Team member who submitted PR: Cristian Urquiza

(Do not email purchasing for updates **before checking** with above team member or uploaded PR. Entire team has access to Team Folder/ PR/ Team Status Doc)

Team-Save your file with team *and* PR number OR FILE WILL BE SENT BACK WITHOUT PARTS ORDERED/NO EXCEPTIONS)

Example of how to SAVE FILE: TEAM_TEST1 _1 (do not need to change here, apply to file name before sending to mentor)

Mentor email: michelle.demers@charlotte.edu

DEPARTMENT: COE Senior Design

VENDOR: Amazon

PROJECT LEAD: Caleb Burton

PROJECT LEAD EMAIL: cburto34@charlotte.edu

PR	VENDOR	RECEIVED DATE	SUBMITTED ORDER TOTAL	ISL PURCHASER	DATE ISL SUBMITTED	ACTUAL ORDER TOTAL	ORDER #
PR_10	Amazon		8.99				

ITEM NO.	QTY	PART NUMBER	PRODUCT DESCRIPTION	UNIT PRICE	TOTAL PRICE	LEAD TIMES	NOTES FOR OFFICE ONLY (DO NOT EDIT)
1	1	B0DQXNB3DN	AGVEE 2 Pack Mid Angle USB4 USB-C Male to USB-C Female Adapter Converter Video USB 4.0 240W 40G Data Extension Coupler Connector, Black	8.99	8.99	5 Days	

VENDOR JUSTIFICATIONS AND QUOTES:
Include vendor name, website LINK, contact person(s) name, email address and phone number. If ordering computer equipment, please provide a reason for why this particular device is needed from this vendor. IT will need to approve all software and computers.
Justification REQUIRED WHEN:
If chosen vendor is not on the Preferred (49er Mart) Vendor list. If P-card is required (see Preferred (49er Mart) Vendor list) for vendors in RED.
QUOTE REQUIRED WHEN:
Item has specific cuts, it's stated on Preferred (49er Mart) Vendor list or it's vendor requested. If vendor is new (not on list) *check with them first*.
Office does not get quotes for teams.

Justification here: "Project X needs item X from vendor X due to the special X that only vendor x can provide." or "Project X requires product X from vendor x as long lead time from other vendors will conflict with project completion."

JUSTIFICATION: FONTAINE_DECAL requires a mid angle USB-C adapter for cable management. Amazon is a preferred vendor. ASIN numbers are listed as "part number" below. https://www.amazon.com/. A similar product from Grainger was available but not to the specification the FONTAINE_DECAL team required. The product did not have the mid angle, and is a USB-C

SIGNOUT FOR PARTS: Team member who fills out PR is responsible for picking up items and noting what is missing OR coordinating with a team member before sending them to sign out for the package. Team members picking up items must make an appointment if not available to receive it during open office hours posted on Canvas and on Purchasing Status Doc. **Availability is not otherwise ensured.**

Figure 42. Purchase Requisite 10 (Amazon)